A Reinforcement Learning–Based EWMA Control Chart with an Application to Urban Mobility Monitoring

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Abstract

Recent studies have applied reinforcement learning to control chart design, but direct learning of the signaling decision remains limited. In this study, we propose a reinforcement learning-based exponentially weighted moving average control chart for process monitoring. The monitoring task is formulated as a Markov decision process in which a proximal policy optimization agent learns a signaling probability from the current sample observation and EWMA statistic. The chart performance is evaluated using the average run length, expected average run length, and extra quadratic loss over a range of shift magnitudes. The numerical results show that the proposed chart generally provides shorter detection delays for small and moderate shifts, and its minimum extra quadratic loss is lower than the corresponding minima for the adaptive EWMA, classical EWMA, and CUSUM charts. We further illustrate the applicability of the approach through an urban mobility monitoring application using bus ridership data from the Chicago Transit Authority. The case study illustrates the signaling patterns of the chart during pronounced variations in observed ridership, including those around 2007–2008 and 2020.

Keywords:
Reinforcement learning; control chart; EWMA; urban mobility; proximal policy optimization; Markov decision process;

1. Introduction

Maintaining high product and process quality in modern manufacturing and service industries is important for meeting customer expectations and minimizing production costs (Montgomery, 2020). As systems become automated, systematic tools to monitor variation are essential to achieve sustainable competitiveness. In practical applications such as transportation systems, raw data often exhibit complex dynamics, including seasonal variations and long-term trends (see Hightower et al. (2024)). To effectively monitor these systems, such nonstationary characteristics are typically removed through preprocessing to isolate a stable, stationary residual series. For monitoring these stationary processes, Zaman et al. (2025) note that adaptive and data-driven schemes can provide more efficient and relevant alerts in practice. A critical challenge in this context is rapidly detecting small to moderate assignable shifts in the process mean. Therefore, for a stationary process with known parameters, reinforcement learning can be used to learn state-dependent detection rules from simulated process trajectories.

Process variation can be broadly classified into two types according to its causes: natural causes and assignable causes (Colosimo et al., 2024). Natural variation arises from the inherent variability of a stable process, including environmental fluctuations and measurement noise. By contrast, assignable causes arise from specific identifiable factors, such as equipment failures, raw material defects, or human errors. These shifts in process characteristics can lead to degradation in the quality of products and services if not detected and corrected promptly. A fundamental objective of quality engineering, therefore, is to distinguish between these two types of variation and respond appropriately to prevent the propagation of faults or unnecessary process stoppages.

Statistical process control (SPC) uses control charts and related statistical methods to detect changes associated with assignable causes. Classical control charts include Shewhart (Shewhart, 1930), cumulative sum (CUSUM) (PAGE, 1954), and exponentially weighted moving average (EWMA) charts (Roberts, 1959). These charts are used to monitor process statis-

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tics over time and raise alarms when significant deviations from the nominal value occur. These charts are designed to balance the trade-off between false alarms, i.e., signaling during natural variation, and detection delay when an actual process shift occurs. In a typical control chart, the in-control (IC) state occurs when the process fluctuates around the target value and remains within the predefined upper and lower control limits (UCL and LCL). In contrast, an assignable cause may induce out-of-control (OOC) behavior, which is detected when the plotted statistic exceeds the control limits. Although effective under ideal conditions, such classical approaches may not fully exploit statistical trends in the data or adapt to evolving dynamics of the monitored process.

In this study, we introduce a control chart based on reinforcement learning (RL) that learns a nonlinear signaling probability from the sample observation and EWMA statistic. This probability is then converted into a signal-or-continue decision. This method offers a more efficient alternative to traditional control charts.

Several studies (Kazmi and Noor-ul Amin, 2025; Abbas et al., 2025; Ajab et al., 2026) have explored the integration of machine learning into SPC to construct data-driven monitoring statistics and adaptive charting rules. A review by Chang et al. (2026) examines applications of artificial intelligence in statistical process monitoring, including control charts, pattern recognition, and anomaly detection. RL provides a framework for learning sequential monitoring decisions from observed states and reward signals. By formulating SPC as a sequential decision-making problem, RL can map the current monitoring state to actions such as selecting observations, adjusting chart parameters, or issuing a signal. The present study differs from existing RL-based control charts. Li et al. (2025) used RL to select the sensor to sample, while Chang and Park (2025) used RL to adjust the moving window length. More recent studies have used RL to select variable sample sizes and sampling intervals in adaptive control charts (Rahimi et al., 2026; Sabahno, 2027). Unlike these studies, the proposed chart uses RL to learn whether to signal or continue monitoring at each sampling time. This approach can learn nonlinear signaling rules, but it requires careful environment setup and policy training.

Our Contribution

We develop an RL-based EWMA control chart (RL-EWMA) for detecting shifts in the mean of a univariate process. We train a signaling agent using the standard Proximal Policy Optimization (PPO) algorithm (Schulman et al., 2017) to learn a monitoring policy that balances detection speed with false alarms.

The key idea is that the agent learns a nonlinear signaling surface from the sample and EWMA statistic. After training, the threshold η is calibrated so that the resulting deterministic signaling rule attains the target in-control average run length. Our main contributions are summarized as follows:

- We formulate the monitoring problem as a sequential stopping decision and train a PPO policy using rewards that encourage prompt detection and discourage false alarms.
- We propose a post-training threshold calibration step that enforces the target IC average run length (ARL) without constraining policy learning. This aligns the final operating point with classical control limit tuning.
- The resulting RL-EWMA chart achieves better OOC detection performance than existing classical charts.

This approach combines the modeling flexibility of a learned nonlinear policy with the statistical rigor of control charts. The proposed chart provides a hybrid monitoring scheme with improved overall performance.

The remainder of the article is organized as follows. Section 2 reviews relevant literature on control charts, RL, and their integration. Section 3 describes the proposed chart, including the MDP formulation and training setup. Section 4 presents the simulation results and comparisons with existing charts. Section 5 discusses additional considerations, including parameter estimation. Section 6 presents the bus ridership case study, and Section 7 concludes the study and outlines future research directions.

2. Literature Review

2.1. Control Charts

SPC techniques primarily employ control charts to monitor process performance and identify unusual deviations. The most fundamental type of control chart is the Shewhart chart, which applies statistically derived limits to indicate when a process statistic (such as the sample mean or range) exceeds a fixed threshold. Shewhart charts are easy to implement and quickly identify large process shifts, but they are comparatively less sensitive to smaller or gradual shifts (Montgomery, 2020). To enhance the ability to detect small, persistent shifts, CUSUM and EWMA charts are commonly applied.

The CUSUM chart accumulates deviations of the process statistic from its target over time and signals an OOC condition when the cumulative sum exceeds a decision threshold. By accumulating deviations, the CUSUM chart can detect small changes in the process much faster than a Shewhart chart (Abbas, 2023). The EWMA statistic assigns greater weight to recent observations, making the chart more sensitive to small, persistent shifts than the Shewhart chart (Ghasemian and Noorossana, 2024). Both CUSUM and EWMA charts rely on specific design parameters that govern their level of sensitivity. These parameters are usually chosen according to statistical criteria and the magnitude of process shifts that are of practical interest. Table 1 summarizes the control charts used for comparison in the current study, together with their plotting statistics, signaling rules,

and design parameters. For the CUSUM chart, k is the reference value and H is the decision interval. For the EWMA chart, λ is the smoothing parameter and L is the control limit multiplier. For the AEWMA chart, $\varphi(\cdot)$ is the adaptive score function and h is the signaling threshold. Here, ARL_0 denotes the IC ARL, i.e., the expected number of observations until a false alarm occurs. Classical control charts use prespecified statistics, parameters, and signaling limits. Adaptive and data-driven approaches allow parts of the monitoring rule to depend on the observed process state (Psarakis, 2015), which motivates the use of RL for sequential monitoring decisions.

2.2. Reinforcement Learning

RL is an area of machine learning that aims to train autonomous agents to make sequential decisions through trial-and-error interactions with an environment (Sutton and Barto, 2018). In the standard RL model, an agent observes the current state of the environment, takes an action, and receives a reward signal that reflects immediate feedback on its action. The environment then transitions to a new state according to the underlying dynamics. This process is repeated until the agent learns a policy, defined as a mapping from states to actions, that maximizes long-term cumulative reward.

Sequential decision-making is modeled as an MDP, characterized by the state space, action space, reward function, and state transition probabilities. RL algorithms can be broadly categorized into value-based methods, policy-based methods, and actor–critic methods (Sutton and Barto, 2018). Value-based methods, such as Q-learning, aim to learn a value function that approximates the expected total reward for each state–action pair. Actions are then selected by choosing those with the highest estimated value. In contrast, policy-based approaches optimize the policy directly with respect to expected reward, while actor–critic methods integrate both ideas by using an explicit policy (the actor) that is guided by a value function (the critic). A central characteristic of RL is the trade-off between exploration and exploitation. The agent must balance exploiting current knowledge to obtain immediate rewards with exploring new actions that might yield higher future rewards. Through repeated interaction with the environment, the agent iteratively improves its policy and learns from experience.

In process monitoring, the state can contain current observations and chart statistics, while the actions determine how monitoring proceeds. The reward function can penalize false alarms and detection delays, allowing the agent to learn a policy for the specified monitoring objective.

2.3. RL-Based Control Charts

The incorporation of RL into SPC has created new possibilities for developing more adaptive and intelligent approaches to process monitoring. Viharos and Jakab (2021) applied RL to SPC in manufacturing. Their approach discretizes process observations into control chart regions and uses a Q-table to guide production control, with the objective of reducing costs while maintaining product quality.

Li et al. (2025) formulated the task as a sequential decision-making problem that an RL agent can learn to solve using Q-learning with a nonparametric CUSUM chart. Their method selects the most informative sensor at each sampling time and uses the CUSUM statistic as part of the system state. The results showed that the RL-based chart enabled faster detection of mean shifts in high-dimensional streaming data than classical fixed-sampling techniques.

Chang and Park (2025) introduced a real-time contrast (RTC) control chart based on RL. In their study, an RL agent is trained to determine the appropriate length of the moving window, namely the number of most recent observations employed to compute the contrast statistic. They used the current process data stream as the state of the environment and defined the action as adjusting the moving window length, either increasing or decreasing it, before the subsequent sample is taken. This RL-based control chart was shown to identify process shifts more quickly than a traditional RTC chart employing a fixed window (Chang and Park, 2025). In this design, RL adjusts the moving window length, while the RTC chart retains the signaling function. Rahimi et al. (2026) proposed an RL framework for adaptive process monitoring with variable sampling strategies. Their framework addresses challenging process conditions, including nonnormality and autocorrelation. Sabahno (2027) used a deep Q-network to select the sample size and sampling interval in an adaptive EWMA chart. The learned policy adjusts sampling effort according to the monitoring state. The classical EWMA statistic and control limits retain the signaling function. These examples from the literature show how RL techniques can improve the performance of control charts.

The integration of RL with SPC is an emerging direction in quality control research. RL provides a framework for continual learning and decision optimization that can address some limitations of classical SPC, such as the need for manual adjustment and the difficulty of handling changing process conditions.

3. Methodology

3.1. The PPO Algorithm

Proximal Policy Optimization (PPO) is an actor–critic policy gradient method that uses a clipped surrogate objective to constrain the size of policy updates, ensuring stable and reliable improvement without requiring complex second-order mathematical calculations (Schulman et al., 2017). To formalize the PPO update, let π_k denote the current policy and π denote the candidate policy. The objective $J(\pi)$ is defined as the expected discounted return when following π, with γ denoting the discount factor. The quantity $p^{\gamma}_{\pi_k}(s)$ is the normalized discounted

Table 1: Control charts used for comparison and their design parameters.

Chart	Plotting Statistic	Signaling Rule	Design Parameters
CUSUM(PAGE, 1954)	$C_t^+ = \max(0, C_{t-1}^+ + X_t - k)$ $C_t^- = \max(0, C_{t-1}^- - X_t - k)$	$\max(C_t^+, C_t^-) > h$	k: reference value h: decision interval tuned to ARL_0
EWMA(Roberts, 1959)	$Z_t = \lambda X_t + (1-\lambda) Z_{t-1}$	$Z_t > \mathrm{UCL}$ or $Z_t < \mathrm{LCL}$ $\mathrm{UCL} = \mu_0 + L\sigma\sqrt{\frac{\lambda}{2-\lambda}}$ $\mathrm{LCL} = \mu_0 - L\sigma\sqrt{\frac{\lambda}{2-\lambda}}$	λ: smoothing parameter L: control limit multiplier tuned to ARL_0
AEWMA (Capizzi and Masarotto, 2003)	$x_t = x_{t-1} + \varphi(e_t),\ e_t = y_t - x_{t-1},\ x_0 = \mu_0$	$\lvert x_t - \mu_0 \rvert > h$	$\varphi(\cdot)$: adaptive score function h: signaling threshold tuned to ARL_0

state visitation distribution induced by the current policy. The functions $Q_{\pi_k}(s,a)$ and $V_{\pi_k}(s)$ are the state–action value function and the state value function, respectively, and their difference $A_{\pi_k}(s,a) = Q_{\pi_k}(s,a) - V_{\pi_k}(s)$ defines the advantage function. The total variation distance between two action distributions p and q is given by $\mathrm{TV}(p,q) = \frac{1}{2}\|p-q\|_1$. The bound on policy improvement can be expressed as:

$$J(\pi) - J(\pi_k) \;\geq\; \frac{1}{1-\gamma}\,\mathbb{E}_{s\sim p^{\gamma}_{\pi_k},\, a\sim\pi_k(\cdot|s)}\left[\frac{\pi(a|s)}{\pi_k(a|s)} A_{\pi_k}(s,a)\right] - \kappa\,\mathbb{E}_{s\sim p^{\gamma}_{\pi_k}}\left[\mathrm{TV}(\pi(\cdot|s), \pi_k(\cdot|s))\right]. \tag{1}$$

where κ is an appropriate constant.

PPO replaces the explicit trust region constraint with a clipped surrogate objective (Schulman et al., 2017). Using the likelihood ratio

$$\rho_k(s,a) = \frac{\pi(a|s)}{\pi_k(a|s)}, \qquad \tilde{\rho}_k(s,a) = \mathrm{clip}(\rho_k(s,a), 1-\epsilon, 1+\epsilon),$$

(where $\mathrm{clip}(x,l,u) = \min(\max(x,l),u)$), PPO updates the policy by maximizing

$$\mathbb{E}_{(s,a)\sim p^{\gamma}_{\pi_k}}\Big[\min\left(\rho_k(s,a)A_{\pi_k}(s,a),\ \tilde{\rho}_k(s,a)A_{\pi_k}(s,a)\right)\Big]. \tag{2}$$

The clipping prevents overly large changes in action probabilities from dominating the update. For simplicity, using ρ as the likelihood ratio and A as the advantage, an equivalent piecewise form is

$$\min(\rho A, \mathrm{clip}(\rho, 1-\epsilon, 1+\epsilon)A) = \min(\rho A,\ g(\epsilon, A)),$$
$$g(\epsilon, A) = \begin{cases} (1+\epsilon)A, & A \geq 0, \\ (1-\epsilon)A, & A < 0. \end{cases}$$

Thus, positive advantage actions cannot be increased by more than a factor $(1+\epsilon)$ and negative advantage actions cannot be decreased by more than a factor $(1-\epsilon)$ in the surrogate objective.

In practice, PPO is implemented using an actor–critic method, where the advantages are estimated via Generalized Advantage Estimation (GAE); see Algorithm 1. The training loss combines (i) a value regression term, (ii) the clipped policy term in (2), and (iii) an entropy bonus to discourage premature collapse to nearly deterministic policies (Schulman et al., 2018, 2017). In Algorithm 1, λ_{TD}, λ_{PG}, and λ_{ent} denote the weights assigned to the temporal-difference (TD) value loss term, the policy gradient (PG) surrogate loss term, and the entropy regularization term, respectively.

Algorithm 1: PPO with GAE

Initialize parameters θ, ϕ and environment state s
for iteration $k = 1, 2, \ldots$ **do**
  $\theta_{\mathrm{old}} \leftarrow \theta$
  $(s_1, a_1, r_1, \ldots, s_T) \leftarrow \textsc{rollout}(s, \pi_{\theta_{\mathrm{old}}})$
  $v_t \leftarrow V_\phi(s_t)$ **for** $t = 1{:}T$
  $(\widehat{A}_{1:T}, y_{1:T}) \leftarrow \mathrm{GAE}(r_{1:T}, v_{1:T}; \gamma, \lambda_{\mathrm{GAE}})$
  for $j = 1{:}M$ **do**
    $\rho_t \leftarrow \dfrac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\mathrm{old}}}(a_t \mid s_t)}$ **for** $t = 1{:}T$
    $\tilde{\rho}_t \leftarrow \textsc{clip}(\rho_t)$ **for** $t = 1{:}T$
    $L(\theta,\phi) \leftarrow \dfrac{1}{T}\sum_{t=1}^{T}\Big[\lambda_{\mathrm{TD}}\,\textsc{clip}((V_\phi(s_t) - y_t)^2) - \lambda_{\mathrm{PG}}\min(\rho_t\widehat{A}_t, \tilde{\rho}_t\widehat{A}_t) - \lambda_{\mathrm{ent}}\,H(\pi_\theta(\cdot \mid s_t))\Big]$
    $\theta \leftarrow \theta - \alpha_{\mathrm{lr}}\nabla_\theta L(\theta,\phi)$
  end for
end for

3.2. The Proposed RL-EWMA Control Chart

The RL-based design of the EWMA control chart is formulated as a sequential stopping problem. The process produces observations X_t and is monitored using an EWMA statistic. At each sampling time, the agent observes both the sample value and the EWMA statistic and then chooses either to continue sampling or to issue a signal. Because the true IC or OOC condition is not directly observable, the underlying process is partially observable, with a binary indicator representing the hidden

Table 2: Important notation and definitions used in the study

Symbol	Definition
A_t	Agent action at time t: $A_t \in \{0: \text{ wait}, 1: \text{ signal}\}$
ARL_0	Average run length (ARL) under IC conditions
ARL_1	Average run length under OOC conditions
CED	Conditional expected delay after a change point
c_s	Cost of sampling when in control
η	Decision threshold on $\pi_\theta(A_t = 1 \mid X_t, Z_t)$ for binary signaling
Z_t	EWMA statistic at time t: $Z_t = \lambda X_t + (1-\lambda)Z_{t-1}$
EARL	Expected average run length, an aggregate ARL measure under a shift distribution
EQL	Extra quadratic loss, an aggregate performance index over shift magnitudes
H_t	Hidden process state: $H_t = 0$ (IC), $H_t = 1$ (OOC)
δ	Magnitude of process mean shift in units of σ
c_d	Penalty for delayed detection: sampling while OOC
P_{FA}	Penalty for false alarms
$Q_{0.10}$	10th percentile of the run length distribution
$Q_{0.90}$	90th percentile of the run length distribution
σ	Process standard deviation
X_t	Raw process observation at time t, $X_t \sim \mathcal{N}(\mu_t, \sigma^2)$
R_{sig}	Reward for correct signaling when process is OOC
R_t	Reward at time t, based on sampling, delay, and signaling accuracy
p	Shift probability during training
τ	Shift time
λ	Smoothing parameter for EWMA, $0 < \lambda < 1$
SDRL	Standard deviation of the run length distribution
π_θ	Stochastic policy parameterized by neural network weights

process state. The agent then learns a stopping rule from the observable monitoring statistics.

Table 2 summarizes the main notation used in the modeling and evaluation of the RL-EWMA control chart.

3.3. State, Actions, and Rewards

At each sampling time t, the environment generates a raw process observation X_t. To incorporate recent process history while smoothing random noise, this observation is updated through the EWMA statistic

$$Z_t = \lambda X_t + (1-\lambda)Z_{t-1}, \quad Z_0 = \mu_0, \quad 0 < \lambda < 1, \tag{3}$$

where $X_t \sim \mathcal{N}(\mu_t, \sigma^2)$ is the process sample. Here, X_t is the current observation, while Z_t combines the current observation with exponentially weighted information from previous observations. Consequently, the observable state provided to the RL agent is given by

$$S_t = (X_t, Z_t)^\top \in \mathbb{R}^2. \tag{4}$$

Although Z_t includes the current observation, the pair (X_t, Z_t) separates immediate and accumulated process information. X_t captures the current departure, while the combination permits recovery of the previous EWMA value as $Z_{t-1} = (Z_t - \lambda X_t)/(1-\lambda)$. This allows the policy to distinguish an isolated extreme observation from a persistent shift producing a similar Z_t. In Table 5, alternative states containing X_t only and another with (X_t, Z_t, Z_{t-1}) are explored. The environment maintains an unobserved binary indicator

$$H_t \in \{0, 1\}, \tag{5}$$

where $H_t = 0$ denotes the IC process with mean $\mu_t = \mu_0$ and $H_t = 1$ denotes the OOC process with mean $\mu_t = \mu_0 + \delta\sigma$. Once a shift occurs, the process remains in the OOC state until the agent raises a signal and then the episode ends. At each time t, the agent chooses an action

$$A_t \in \{0, 1\}, \tag{6}$$

where $A_t = 0$ denotes waiting and continuing to sample, whereas $A_t = 1$ denotes signaling an OOC condition and stopping.

Shift occurrences are modeled using a geometric mechanism. While the process is IC ($H_t = 0$), a shift arrives with probability

$$\mathbb{P}(H_{t+1} = 1 \mid H_t = 0) = p. \tag{7}$$

The value of p should be selected so that the agent observes enough IC and OOC episodes to learn a meaningful policy. The EWMA and sample values then induce the transition in the observable state (X_t, Z_t).

The immediate reward function $R(S_t, A_t, H_t)$ is defined as follows:

$$R_t = \begin{cases} -c_s, & A_t = 0,\ H_t = 0, \\ -c_s - c_d, & A_t = 0,\ H_t = 1, \\ R_{\text{sig}}, & A_t = 1,\ H_t = 1, \\ -P_{\text{FA}}, & A_t = 1,\ H_t = 0, \end{cases} \tag{8}$$

where c_s denotes the sampling cost, c_d represents the cost of operating in a defective state, R_{sig} represents the reward for a correct signal, and P_{FA} is the penalty for a false alarm. The reward parameters are used only to shape policy learning and are not interpreted as operational costs. The final chart is calibrated separately through η to attain the target ARL_0. Therefore, the present design is statistical, not economic, and the reward parameters have no direct mapping to ARL, EQL, or EARL. The reward values are normalized learning parameters. We set $c_s = 0$ because sampling occurs at every monitoring step and sampling frequency is not optimized. The correct signal reward is fixed at $R_{\text{sig}} = 1$ as the reference scale, while $P_{\text{FA}} = 1$ gives equal nominal weight to false alarms and correct signals. The delay penalty $c_d = 0.20$ imposes a smaller recurring penalty for each delayed signal. In practice, these parameters may be adjusted to reflect the relative importance of false alarms and detection delay. Sub-

stituting the selected parameter values into the reward function gives

$$R_t = \begin{cases} 0, & A_t = 0,\ H_t = 0, \\ -0.20, & A_t = 0,\ H_t = 1, \\ 1, & A_t = 1,\ H_t = 1, \\ -1, & A_t = 1,\ H_t = 0. \end{cases} \tag{9}$$

Hence, every additional waiting decision after a shift receives a penalty of -0.20, while an incorrect signal under IC conditions receives a penalty of -1. This structure encourages faster signaling after a shift while discouraging false signals.

Figure 1 shows the RL control loop used in this study. At each time t, the PPO agent observes S_t, selects an action A_t, and the control chart environment returns the next state S_{t+1} and reward.

3.4. Evaluation Metrics

We evaluate the proposed chart using the following performance metrics: ARL, conditional expected delay (CED), expected ARL (EARL), and extra quadratic loss (EQL). The ARL is the expected number of observations until a signal is triggered. Two variants are commonly reported. The first is ARL_0, which is the expected run length when the process is IC, i.e., when no shift occurs. Larger ARL_0 values indicate greater resistance to false alarms. The second is $\mathrm{ARL}_1(\delta)$, which is the expected run length when a shift of magnitude δ occurs. Smaller $\mathrm{ARL}_1(\delta)$ values indicate faster detection of OOC conditions. The ARL values are computed separately for each shift magnitude δ in the test set, providing information about the chart's sensitivity to different process shifts.

In addition to ARL, CED is used to evaluate detection performance when the process shift occurs after the chart has already been operating for some time. For a change point τ, CED is defined as the expected detection delay conditional on no false alarm before τ (Kenett and Pollak, 2012):

$$\mathrm{CED}_\tau = \mathbb{E}[(T - \tau + 1) \mid T \geq \tau],$$

where T is the signal time. In simulations, if K trajectories are generated and $N_\tau = \sum_{i=1}^{K} \mathbf{1}(T_i \geq \tau)$ of them survive without a false alarm before τ, then CED is estimated by

$$\widehat{\mathrm{CED}}_\tau = \frac{1}{N_\tau} \sum_{i:T_i \geq \tau} (T_i - \tau + 1).$$

The EQL metric provides an aggregate measure of chart performance across multiple shift magnitudes and is defined as follows:

$$\mathrm{EQL} = \frac{1}{\delta_{\max} - \delta_{\min}} \int_{\delta_{\min}}^{\delta_{\max}} \delta^2 \, \mathrm{ARL}_1(\delta)\, d\delta. \tag{10}$$

This formulation emphasizes detection performance across a range of shifts in the process mean. The evaluation grid and numerical approximation are specified in the experimental setup.

To further summarize performance over a range of possible shift sizes, we also report EARL. This metric averages $\mathrm{ARL}_1(\delta)$ over an assumed shift distribution. Following the use of random shift size distributions in control chart evaluation, EARL is computed with a prespecified shift size density when the future shift magnitude is unknown (Chen and Chen, 2007; Huang et al., 2016). Here, an exponential density is used as a weighting function to give more importance to small and moderate shifts, which are usually more difficult to detect. The EARL is computed as

$$\mathrm{EARL} = \int_0^\infty \mathrm{ARL}_1(\delta) f(\delta)\, d\delta,$$

where $f(\delta) = \beta^{-1} \exp(-\delta/\beta)$ for $\delta \geq 0$ is the exponential density of the shift magnitude. The grid and scale parameter are specified in the experimental setup. Lower EARL values indicate better overall detection performance under the assumed shift distribution.

3.5. Training Process of the Proposed Chart

The RL agent is trained once and then converted into a deterministic control chart using a probability threshold. Therefore, training and evaluation use different shift mechanisms but the same learned policy.

During training, the environment operates under the geometric shift mechanism outlined previously. While $H_t = 0$, a Bernoulli trial with probability p determines whether a shift should be introduced. The shift magnitudes δ are sampled independently from a uniform distribution, $\delta \sim \mathcal{U}(0.50, 3.00)$, so the agent sees a mixture of small and large shifts. The agent observes X_t and Z_t, selects $A_t \in \{0, 1\}$, and receives the corresponding reward. A neural network policy π_θ is trained using the PPO algorithm to maximize the expected discounted return. The policy used in the numerical study was the final checkpoint obtained after the prespecified 200,000 training steps. For any X_t and Z_t, the retained policy provides the probability

$$p_t = \pi_\theta(A_t = 1 \mid X_t, Z_t),$$

which is interpreted as the probability assigned to the signal action at time t. The learning curves and optimization diagnostics in Figure C.13 in the Appendix show the stability of PPO training.

The hardware, software, training time, and policy details are reported in Appendix Appendix D.

For performance evaluation, the environment is switched to a deterministic mode to match standard SPC practice. For a given shift magnitude δ and shift time τ:

- For IC experiments ($\delta = 0$), the process mean remains at μ_0 throughout the run.

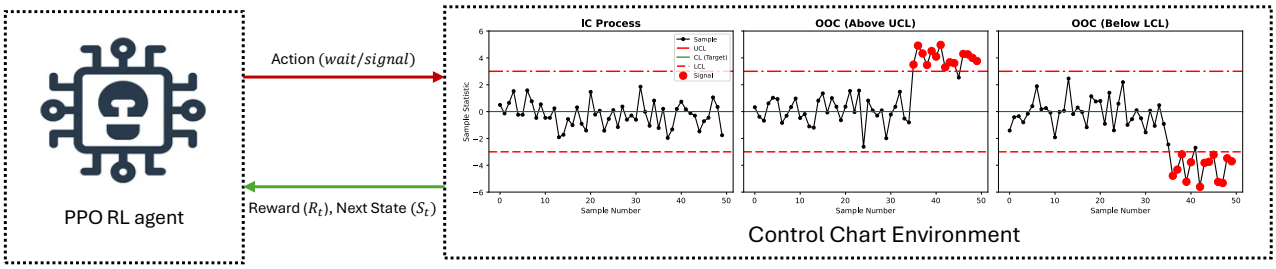


Figure 1: Interaction between the PPO agent and the control chart environment, showing state transitions ($S_t \to S_{t+1}$), rewards ($R_t \to R_{t+1}$), and the agent's action A_t.

- For OOC experiments ($\delta > 0$), the mean is μ_0 for $t < \tau$ and $\mu_0 + \delta\sigma$ for $t \geq \tau$.

This setup enables a direct comparison between the RL-EWMA chart and existing classical control charts.

The neural network policy is stochastic. To obtain a deterministic chart comparable to a classical EWMA scheme, the probability p_t is converted to a binary decision using a threshold $\eta \in [0, 1]$:

$$d_\eta(X_t, Z_t) = \begin{cases} 1, & \pi_\theta(A_t = 1 \mid X_t,\ Z_t) \geq \eta, \\ 0, & \pi_\theta(A_t = 1 \mid X_t,\ Z_t) < \eta, \end{cases} \tag{11}$$

where $d_\eta(X_t,\ Z_t) = 1$ indicates that a signal occurs. The IC ARL and conditional OOC detection delay are given by

$$\mathrm{ARL}_0(\eta) = \mathbb{E}_0[T_\eta], \qquad \mathrm{CED}_\tau(\eta; \delta) = \mathbb{E}_{\delta,\tau}[T_\eta - \tau + 1 \,\big|\, T_\eta \geq \tau].$$

To place the RL chart on the same footing as a classical EWMA, the threshold η is calibrated so that $\mathrm{ARL}_0(\eta)$ matches a prescribed design value $\mathrm{ARL}_0^\star$ (for example, 500). Empirically, $\mathrm{ARL}_0(\eta)$ is monotone increasing in η, so a simple bisection search is sufficient. Starting with $\eta_L = 0$ and $\eta_U = 1$, the trial values $\eta = (\eta_L + \eta_U)/2$ are tested by Monte Carlo simulation of IC trajectories, and the interval is adjusted until $\mathrm{ARL}_0(\eta)$ is within a tolerance of $\mathrm{ARL}_0^\star$.

Once the calibrated value $\eta^\star$ is obtained, all reported values of ARL_0 and ARL_1 for the RL chart are based on the deterministic rule (11) with $\eta = \eta^\star$.

For completeness, we summarize the RL training procedure and ARL evaluation with threshold calibration in Algorithms 2 and 3, respectively.

Algorithm 2: Agent training using PPO

Inputs : Smoothing parameter λ, reward parameters $(c_s, c_d, R_{\mathrm{sig}}, P_{\mathrm{FA}})$, PPO hyperparameters;
Outputs : Trained policy π_θ;
Initialization: Policy network π_θ, value network V_ϕ, and optimizer;

Set shift probability p;
for $iter = 1, \ldots, N_{train}$ **do**
  Reset the environment to IC, set $Z_0 = \mu_0$ and $H_0 = 0$;
  for $t = 1, \ldots, T_{rollout}$ **do**
    if $H_t = 0$ **then**
      Generate a shift with probability p;
      if *a shift occurs* **then**
        $H_t \leftarrow 1$ and fix the shifted mean $\mu_0 + \delta\sigma$ for the rest of the episode;
    Sample $X_t \sim \mathcal{N}(\mu(H_t), \sigma^2)$ and update $Z_t = \lambda X_t + (1 - \lambda) Z_{t-1}$;
    Sample action A_t from the current policy $A_t \sim \pi_\theta(\cdot \mid X_t,\ Z_t)$;
    Compute reward R_t from the cost structure above;
    if $A_t = 1$ **then**
      Terminate the episode;
    Store (S_t, A_t, R_t, S_{t+1}) in a trajectory buffer;
  From the collected batch, compute advantage estimates and the PPO objective;
  Update θ and ϕ by several epochs of stochastic gradient steps on the PPO objectives.

4. Results and Discussion

In this section, we present the experimental setup and numerical results. We also discuss comparative performance against existing methods.

4.1. Experimental Setup

We simulate a Gaussian process with constant variance ($\sigma^2 = 1$) and mean shifts introduced at change point τ. The observations follow $X_t \sim \mathcal{N}(0, 1)$ under the IC condition and $X_t \sim \mathcal{N}(\delta, 1)$ after the shift occurs. We evaluate the chart over the shift magnitudes $\delta \in$

Algorithm 3: Threshold calibration and ARL evaluation

Inputs : Trained policy π_θ, target $\mathrm{ARL}_0^\star$, tolerance ε, maximum bisection iterations $K_{\max}$;
Outputs : Calibrated threshold $\eta^\star$, ARL_0 and ARL_1 estimates for the RL-EWMA chart;

Calibration of $\eta^\star$:;
Initialize $\eta_L = 0$, $\eta_U = 1$;
for $k = 1, \ldots, K_{\max}$ **do**
  Set $\eta_k = (\eta_L + \eta_U)/2$;
  Simulate many IC trajectories using the deterministic rule $d_{\eta_k}(X_t, Z_t)$ in (11); record the stopping times T_{η_k} and compute $\widehat{\mathrm{ARL}}_0(\eta_k)$ as their sample mean;
  if $\left|\widehat{\mathrm{ARL}}_0(\eta_k) - \mathrm{ARL}_0^\star\right| \le \varepsilon$ **then**
    Set $\eta^\star = \eta_k$ and stop;
  if $\widehat{\mathrm{ARL}}_0(\eta_k) < \mathrm{ARL}_0^\star$ **then**
    Set $\eta_L = \eta_k$;
  else
    Set $\eta_U = \eta_k$;
ARL **evaluation with $\eta^\star$:**;
for *each shift magnitude δ and shift time τ of interest* **do**
  Simulate K trajectories with deterministic mean change at τ and charting rule $d_{\eta^\star}(X_t, Z_t)$;
  if $\delta = 0$ *(IC)* **then**
    Record all run lengths $T_{\eta^\star}$ and compute ARL_0 as their average;
  else
    Retain only runs with $T_{\eta^\star} \ge \tau$ and record detection delays $T_{\eta^\star} - \tau + 1$; compute ARL_1 as their mean.

$\{0.25, 0.50, 0.75, 1.00, 1.50, 2.00, 2.50, 3.00, 3.50, 4.00, 5.00, 6.00\}$. For the EARL calculation, the exponential scale parameter was set to the mean of the complete grid, $\beta = 30/13 \approx 2.3077$. The estimated EARL was computed as

$$\widehat{\mathrm{EARL}} = \sum_{\delta_j \in \mathcal{D}_0} \left[\frac{\exp(-\delta_j/\beta)}{\sum_{\delta_k \in \mathcal{D}_0} \exp(-\delta_k/\beta)} \right] \widehat{\mathrm{ARL}}(\delta_j),$$

where $\mathcal{D}_0$ consists of $\delta = 0$ together with the OOC evaluation grid stated above. The distribution was truncated at $\delta = 6.00$, and the weights were normalized over the complete evaluation grid.

For the EQL calculation, $\mathcal{D}$ denotes the OOC evaluation grid stated above. Letting $g(\delta_j) = \delta_j^2 \widehat{\mathrm{ARL}}_1(\delta_j)$, the estimated EQL was computed as

$$\widehat{\mathrm{EQL}} = \frac{1}{\delta_m - \delta_1} \sum_{j=1}^{m-1} \frac{\delta_{j+1} - \delta_j}{2} \left[g(\delta_j) + g(\delta_{j+1}) \right],$$

where $\delta_1 = 0.25$, $\delta_m = 6.00$, and $m = 12$. Thus, the calculation was restricted to $[0.25, 6.00]$, with no extrapolation.

We conducted the evaluation using 10,000 trajectories per configuration to reduce Monte Carlo variability. The main policy was trained using $\delta \sim \mathcal{U}(0.50, 3.00)$. A wider training range covering $\delta \in [0.25, 6.00]$ was examined separately in the sensitivity analysis.

To represent different levels of EWMA memory, we evaluated $\lambda \in \{0.05, 0.14, 0.25, 0.50, 0.70\}$. Similar values were used in previous studies (Capizzi and Masarotto, 2003; Jabeen et al., 2022; Abbas, 2023). Smaller values provide stronger smoothing and longer memory, while larger values place more weight on recent observations. For every combination of λ and δ, we simulated multiple episodes to estimate ARL_0 and ARL_1. The shifts were introduced at $\tau = 1$, so the reported values of ARL_1 correspond to the zero-state case.

The performance of the proposed chart is evaluated against CUSUM as well as classical and adaptive EWMA charts, all configured to attain $\mathrm{ARL}_0 \approx 100$ in the first scenario and $\mathrm{ARL}_0 \approx 500$ in the second.

The AEWMA results were reproduced from Capizzi and Masarotto (2003). For each selected EWMA smoothing parameter λ and CUSUM reference value k, the control limit multiplier L and decision interval H were calibrated using the `spc` package in R to attain the target ARL_0 (Knoth, 2025). The PPO hyperparameters, environment settings, and reward values used to generate the proposed chart's numerical results are reported in Table A.9 in the Appendix.

4.2. Comparative Analysis

This section compares the proposed RL-EWMA chart with the AEWMA chart (Capizzi and Masarotto, 2003), the classical EWMA chart, and CUSUM charts for $\mathrm{ARL}_0 \approx 100$ and $\mathrm{ARL}_0 \approx 500$. Because ARL_0 is calibrated to be approximately equal across all charts, the comparison focuses on (i) the variation in ARL_1 across shift magnitudes and (ii) the aggregate EQL and EARL metrics. The main results are summarized as follows:

- For $\mathrm{ARL}_0 \approx 100$ (Table 3), the RL-EWMA generally yields smaller ARL_1 values than AEWMA, the classical EWMA, and CUSUM for small to moderate shifts ($\delta \in \{0.25, 0.50, 0.75, 1.00, 1.50\}$). For example, at $\delta = 0.50$, the ARL_1 values for the RL-EWMA chart range from 13.12 to 20.37 for different λ values, while the AEWMA values range from 17.64 to 20.30, the classical EWMA values range from 17.41 to 36.60, and the CUSUM values range from 19.33 to 48.83 depending on the reference parameter.

Table 3: ARL comparison of RL-EWMA, AEWMA, EWMA, and CUSUM charts with $ARL_0 \approx 100$.

	RL-EWMA					AEWMA			EWMA					CUSUM				
δ	$\lambda = 0.05$	$\lambda = 0.14$	$\lambda = 0.25$	$\lambda = 0.50$	$\lambda = 0.70$	φ_{hu}	φ_{bs}	φ_{cb}	$\lambda = 0.05$	$\lambda = 0.14$	$\lambda = 0.25$	$\lambda = 0.50$	$\lambda = 0.70$	$k = 0.25$	$k = 0.50$	$k = 0.75$	$k = 1.00$	$k = 2.00$
0.00	103.68	102.57	100.32	103.53	99.22	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
0.25	**26.63**	30.30	31.92	41.77	41.83	41.38	40.53	46.53	38.80	44.13	50.77	63.15	71.08	44.54	52.66	60.86	67.86	80.21
0.50	**13.12**	13.94	14.40	18.62	20.37	20.30	17.64	18.96	17.41	18.18	20.84	28.91	36.60	19.33	21.70	26.60	32.55	48.83
0.75	**8.17**	**8.17**	8.26	10.33	10.90	13.01	10.59	10.41	10.82	10.30	11.02	14.66	19.15	11.57	11.46	13.28	16.25	27.94
1.00	5.94	5.76	**5.66**	6.43	6.83	9.38	7.43	6.92	7.83	7.02	7.06	8.60	10.99	8.19	7.39	7.90	9.21	16.39
1.50	3.62	3.45	**3.33**	3.41	3.48	5.67	4.52	4.10	5.09	4.28	3.99	4.14	4.77	5.19	4.25	4.04	4.21	6.57
2.00	2.50	2.40	2.38	2.29	**2.27**	3.68	3.15	2.95	3.82	3.12	2.81	2.62	2.74	3.84	3.01	2.70	2.61	3.30
2.50	1.86	1.82	1.87	1.73	**1.72**	2.46	2.37	2.35	3.09	2.50	2.21	1.93	1.89	3.08	2.38	2.05	1.90	2.03
3.00	1.45	1.47	1.53	1.41	**1.40**	1.73	1.87	1.99	2.61	2.13	1.85	1.54	1.46	2.59	2.00	1.67	1.52	1.47
3.50	1.23	1.23	1.30	**1.21**	**1.21**	1.34	1.53	1.74	2.28	1.88	1.58	1.30	1.23	2.26	1.73	1.41	1.28	**1.21**
4.00	1.09	1.10	1.14	1.10	1.09	1.14	1.30	1.52	2.08	1.68	1.37	1.14	1.10	2.06	1.51	1.22	1.13	**1.08**
5.00	**1.01**	**1.01**	1.02	**1.01**	**1.01**	1.02	1.06	1.17	1.85	1.28	1.09	1.02	**1.01**	1.80	1.16	1.04	1.02	**1.01**
6.00	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	1.01	1.02	1.51	1.06	1.01	**1.00**	**1.00**	1.44	1.02	**1.00**	**1.00**	**1.00**
EQL	15.59	15.55	15.79	**15.54**	15.57	17.66	17.84	18.94	26.60	20.30	17.87	16.62	16.86	26.18	19.04	17.06	16.68	18.14
EARL	**24.55**	24.97	24.90	27.69	27.34	28.25	27.17	28.11	27.20	27.71	29.01	32.38	35.34	28.41	29.54	31.53	33.78	40.03

Table 4: ARL comparison of RL-EWMA, AEWMA, EWMA, and CUSUM charts with $ARL_0 \approx 500$.

	RL-EWMA					AEWMA			EWMA					CUSUM				
δ	$\lambda = 0.05$	$\lambda = 0.14$	$\lambda = 0.25$	$\lambda = 0.50$	$\lambda = 0.70$	φ_{hu}	φ_{bs}	φ_{cb}	$\lambda = 0.05$	$\lambda = 0.14$	$\lambda = 0.25$	$\lambda = 0.50$	$\lambda = 0.70$	$k = 0.25$	$k = 0.50$	$k = 0.75$	$k = 1.00$	$k = 2.00$
0.00	507.45	506.90	508.34	498.15	505.10	500.00	500.00	500.00	500.00	500.00	500.00	500.00	500.00	500.00	500.00	500.00	500.00	500.00
0.25	**61.64**	81.21	104.91	143.52	163.93	98.51	105.87	119.45	84.01	124.48	170.34	254.83	308.89	94.80	145.53	200.53	249.52	362.50
0.50	**24.00**	26.70	34.07	51.32	62.93	40.94	33.70	33.76	28.76	34.96	48.30	88.81	129.16	31.08	38.87	57.04	81.44	183.68
0.75	14.09	**13.78**	15.54	22.53	27.76	25.04	17.95	16.16	16.37	16.44	20.12	35.92	56.20	17.54	17.32	22.12	30.90	87.88
1.00	9.79	**8.82**	9.44	11.95	14.53	17.59	11.78	10.16	11.38	10.21	11.14	17.48	27.29	12.17	10.52	11.60	14.67	43.50
1.50	6.00	5.07	**4.81**	5.08	5.65	10.11	6.50	5.77	7.11	5.73	5.46	6.53	8.97	7.58	5.82	5.45	5.75	12.90
2.00	4.21	3.55	3.23	**3.10**	3.22	6.08	4.16	4.06	5.22	4.03	3.61	3.63	4.26	5.55	4.06	3.55	3.41	5.21
2.50	3.15	2.73	2.46	2.26	**2.23**	3.66	2.89	3.15	4.17	3.15	2.74	2.50	2.61	4.41	3.15	2.67	2.45	2.81
3.00	2.43	2.18	1.99	1.80	**1.79**	2.29	2.14	2.56	3.50	2.61	2.26	1.93	1.86	3.69	2.60	2.18	1.94	1.87
3.50	1.92	1.82	1.67	1.51	1.51	1.60	1.67	2.10	3.04	2.26	1.95	1.58	1.47	3.19	2.25	1.87	1.62	**1.43**
4.00	1.54	1.54	1.44	1.40	1.29	1.26	1.37	1.75	2.69	2.04	1.73	1.34	1.24	2.84	2.03	1.63	1.38	**1.20**
5.00	1.13	1.16	1.12	1.06	1.06	1.04	1.08	1.27	2.16	1.75	1.32	1.07	1.04	2.26	1.72	1.24	1.09	**1.03**
6.00	1.02	1.02	1.01	1.01	**1.00**	**1.00**	1.01	1.05	2.00	1.37	1.07	1.01	**1.00**	2.02	1.33	1.04	1.01	**1.00**
EQL	20.94	20.12	19.28	**18.85**	19.41	22.19	20.17	22.54	34.29	26.46	22.33	21.03	22.89	35.86	26.30	21.69	20.60	26.44
EARL	**97.58**	100.37	105.18	112.41	118.91	106.46	104.68	106.27	101.03	107.21	115.97	135.96	152.63	103.20	110.90	121.73	133.28	173.22

Note: Bold values indicate the minimum ARL in each row.

For $\delta = 1.00$, the RL-EWMA ARL_1 ranges from 5.66 to 6.83, whereas the AEWMA achieves values between 6.92 and 9.38, the classical EWMA between 7.02 and 10.99, and the CUSUM between 7.39 and 16.39. For larger shifts ($\delta \geq 3.00$), all charts approach $ARL_1 \approx 1$, and the differences in performance are negligible.

- In terms of aggregate performance, the proposed chart achieves lower EQL values (15.54–15.79) than the AEWMA (17.66–18.94), classical EWMA (16.62–26.60), and CUSUM (16.68–26.18) charts, indicating better overall detection performance.

- For $ARL_0 \approx 500$ (Table 4), similar trends are observed. For small shifts, the best-performing RL-EWMA design yields lower ARL_1 values than the best-performing configurations of the competing charts. For instance, at $\delta = 0.25$, RL-EWMA ARL_1 values range from 61.64 to 163.93, while AEWMA ranges from 98.51 to 119.45, classical EWMA from 84.01 to 308.89, and CUSUM from 94.80 to 362.50. At $\delta = 0.50$, RL-EWMA ARL_1 ranges from 24.00 to 62.93, compared with AEWMA values between 33.70 and 40.94, classical EWMA values between 28.76 and 129.16, and CUSUM values between 31.08 and 183.68. For moderate shifts ($\delta \in [1.00, 2.50]$), the RL-EWMA remains competitive and frequently outperforms the other charts, while for larger shifts ($\delta \geq 4.00$) all charts again converge to $ARL_1 \approx 1$.

- For $ARL_0 \approx 500$, the minimum EQL obtained by RL-EWMA is 18.85, compared with 20.17 for AEWMA, 21.03 for EWMA, and 20.60 for CUSUM. However, the ranges overlap across some designs. The minimum occurs for the RL-EWMA chart with $\lambda = 0.50$.

- The EARL values offer an additional perspective on chart performance when the shift magnitude is modeled as a random variable. For $ARL_0 \approx 100$, the minimum EARL is achieved by the RL-EWMA chart with $\lambda = 0.05$, yielding an EARL of 24.55. This suggests that, under the assumed exponential distribution of shifts, the proposed chart provides the most effective overall detection capability across likely shift magnitudes. Similarly, for $ARL_0 \approx 500$, the RL-EWMA chart with $\lambda = 0.05$ again attains the lowest EARL, equal to 97.58. Overall, although the optimal EQL values occur at $\lambda = 0.50$, the EARL metric favors a smaller smoothing parameter, because the exponential shift distribution assigns greater weight to smaller shifts.

For visualization, we selected RL-EWMA with $\lambda = 0.50$, AEWMA with φ_{bs}, EWMA with $\lambda = 0.50$, and CUSUM with $k = 1.00$. These choices are based on the lowest mean EQL across the two target ARL_0 levels. A log scale was used in Figure 2 to improve readability because the ARL values vary widely across shift magnitudes. Comparisons with the AEWMA, classical EWMA, and CUSUM charts show that suitable RL-EWMA configurations are competitive for detecting small to moderate shifts and attain the lowest overall EQL values.

For further comparison, a deep Q-network (DQN) and a logistic regression classifier were implemented using the same state variables. DQN learned action values for the Continue and Signal actions, while logistic regression learned a linear signaling rule. Each method was calibrated separately to $ARL_0 = 500$.

Table 5 shows that the baseline PPO policy using (X_t, Z_t) has lower ARL values for small shifts and performs similarly to DQN for larger shifts. The RL-based methods also generally outperform the linear classifier. Table 5 also compares the learning methods and PPO state representations. Using X_t alone produces substantially longer run lengths for small and moderate shifts. Adding X_t to Z_t reduces the ARL relative to using Z_t alone for shifts from $\delta = 0.25$ to 2.00. Including Z_{t-1} provides modest additional reductions for the smallest shifts but does not consistently improve performance at larger shifts. These results support (X_t, Z_t) as a compact state containing both immediate and accumulated process information.

4.3. Sensitivity Analysis

A sensitivity analysis was conducted for the target $ARL_0 = 500$. The PPO agent was implemented using Stable-Baselines3, which provides reliable implementations of RL algorithms with commonly used hyperparameter settings. The analysis focused on problem-specific factors that directly influence the learned signaling policy. The shift probability, delay penalty, training shift distribution, network architecture, and discount factor were varied, while all remaining settings were held at their baseline values. Each trained policy was calibrated separately to the target ARL_0. Table 6 shows that the ARL, EQL, and EARL results are generally stable across the tested settings. The smaller network and $p = 0.01$ yield slightly higher ARL values for some small shifts, whereas the larger network and $p = 0.05$ provide modest improvements. Expanding the training range to $\delta \sim \mathcal{U}(0.25, 6.00)$ produces similar results. A higher delay penalty yields slightly lower ARL values for small shifts, whereas a lower penalty yields lower ARL values for larger shifts. The baseline value $c_d = 0.20$ provides balanced ARL performance across the shift range.

During model development, training was repeated several times using the same fixed settings, and the calibrated policies produced similar ARL profiles. The 200,000 training steps allowed the learned policy to stabilize, while post-training threshold calibration aligned each policy with the same target ARL_0.

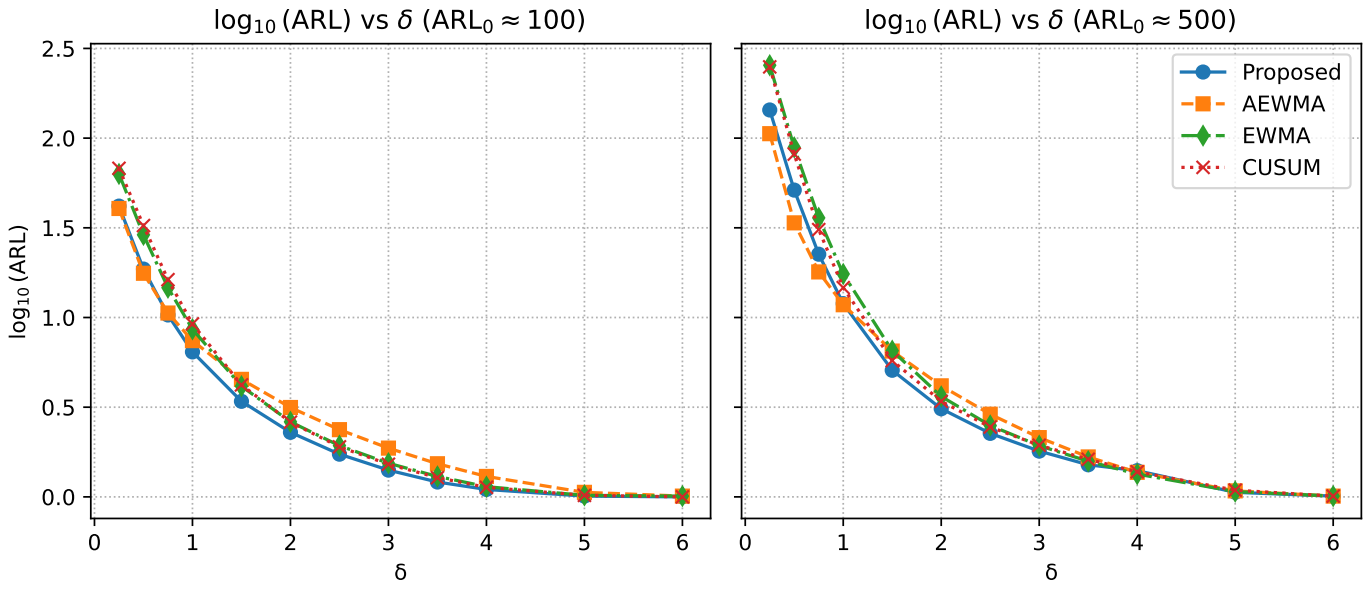


Figure 2: $\log_{10}(\text{ARL})$ versus shift magnitude δ for $\text{ARL}_0 \approx 100$ and $\text{ARL}_0 \approx 500$, comparing the proposed RL-EWMA chart with AEWMA, classical EWMA, and CUSUM charts.

Table 5: Comparison of learning methods and PPO state representations for the target $\text{ARL}_0 \approx 500$.

Method and state	ARL(δ)												
	0.00	0.25	0.50	0.75	1.00	1.50	2.00	2.50	3.00	3.50	4.00	5.00	6.00
DQN, (X_t, Z_t)	498.76	149.36	58.49	25.25	11.91	5.04	3.12	2.32	1.94	1.48	1.35	1.05	1.01
Linear classifier, (X_t, Z_t)	497.24	162.90	63.51	27.04	12.26	7.00	3.57	2.46	2.13	1.97	1.86	1.52	1.17
PPO, X_t	502.58	228.52	113.83	60.13	32.70	11.71	5.21	2.83	1.83	**1.35**	**1.15**	**1.02**	**1.00**
PPO, Z_t	505.94	150.99	54.77	24.61	12.72	5.37	3.19	**2.26**	**1.77**	1.44	1.25	1.04	**1.00**
PPO, (X_t, Z_t)	498.15	143.52	51.32	22.53	11.95	5.08	**3.10**	**2.26**	1.80	1.51	1.40	1.06	1.01
PPO, (X_t, Z_t, Z_{t-1})	500.06	**139.27**	**48.04**	**21.09**	**11.30**	**5.01**	3.12	2.36	2.01	1.76	1.55	1.19	1.03

Note: Bold values indicate the minimum OOC ARL in each column.

5. Additional Considerations and Extensions

The results presented thus far focus on the primary design and evaluation of the proposed RL-EWMA chart under the assumption that process observations are normally distributed with known parameters and that any shift occurs at the start of monitoring (the zero-state case). This section examines additional considerations that are important for practical implementation, including parameter estimation, the normality assumption, and steady-state ARL performance.

5.1. Effect of Parameter Estimation

Control chart designs usually assume that the IC process parameters, namely the mean μ_0 and standard deviation σ_0, are known in advance, a situation commonly known as Case K. Under this assumption, the chart parameters are calibrated to achieve a specified IC ARL_0. In practice, μ_0 and σ_0 are seldom known and must be estimated from a Phase I reference dataset; this is known as Case U. However, replacing (μ_0, σ_0) with their estimates $(\hat{\mu}_0, \hat{\sigma}_0)$ adds variability to Phase II monitoring and alters both IC and OOC run length performance (Jensen et al., 2006; Castagliola et al., 2016). For EWMA charts, Jones et al. (2001) provide a detailed analysis of the effect of parameter estimation.

To incorporate the effect of parameter estimation, a two-stage approach is used. In Phase I, an IC reference sample is collected and used to estimate μ_0 and σ_0. In Phase II, the control chart is applied using these estimates, and the run length is recorded conditional on the Phase I outcome. Jones et al. (2001) consider several unbiased estimators of σ_0, including $\bar{R}/d_2$, $\bar{S}/c_4$,

Table 6: Sensitivity of the RL-EWMA chart to selected training settings for the target $ARL_0 = 500$.

Factor	Setting	ARL													EQL	EARL
		0.00	0.25	0.50	0.75	1.00	1.50	2.00	2.50	3.00	3.50	4.00	5.00	6.00		
Baseline	$\lambda = 0.50$	498.15	143.52	51.32	22.53	11.95	5.08	3.10	2.26	1.80	1.51	1.40	1.06	1.01	18.85	112.41
Delay penalty	$c_d = 0.05$	504.38	147.91	54.72	24.24	12.52	5.32	3.15	2.24	1.76	1.46	1.25	1.04	1.00	18.75	114.76
Delay penalty	$c_d = 0.50$	504.68	140.66	48.53	21.55	11.30	5.06	3.15	2.36	1.99	1.77	1.55	1.20	1.03	20.47	112.54
Discount factor	$\gamma = 0.95$	500.54	151.85	53.06	24.51	11.23	4.97	3.02	2.28	1.82	1.47	1.39	1.06	1.00	19.01	114.37
Discount factor	$\gamma = 0.995$	500.26	145.72	51.99	23.02	11.46	4.93	3.15	2.39	1.79	1.55	1.43	1.08	1.01	19.33	113.16
Shift probability	$p = 0.01$	499.30	148.06	54.17	23.67	12.01	5.13	3.35	2.34	1.87	1.51	1.42	1.06	1.02	19.43	113.79
Shift probability	$p = 0.05$	497.92	142.66	51.02	22.00	11.25	4.98	3.11	2.28	1.80	1.49	1.36	1.03	1.00	18.76	112.07
Network architecture	$[32, 32]$	498.04	150.41	53.00	25.78	12.22	5.19	3.17	2.31	1.83	1.50	1.41	1.06	1.01	19.32	114.03
Network architecture	$[128, 128]$	500.07	141.65	50.44	22.10	11.94	5.08	3.02	2.28	1.75	1.48	1.40	1.04	1.00	18.86	112.28
Training shifts	$U(0.25, 6.00)$	499.01	140.93	51.40	22.78	12.00	5.12	3.11	2.31	1.85	1.50	1.32	1.01	1.00	18.75	112.23

Note: The baseline settings are $\lambda = 0.50$, $\gamma = 0.99$, $p = 0.02$, $c_d = 0.2$, two hidden layers of 64 neurons, and $\delta \sim \mathcal{U}(0.50, 3.00)$. Only the stated factor was changed in each sensitivity setting.

and the pooled estimator $S_p/c_{4,m}$, and adopt the latter because of its efficiency. Here, we focus on the individual-observation case, which is the analog of their method when the subgroup size equals one. This approach was also adopted by Atalay et al. (2020). For a Phase I reference sample of size m, we estimate

$$\hat{\mu}_0 = \frac{1}{m}\sum_{i=1}^{m} X_i,$$

$$\hat{\sigma}_0 = \frac{s}{c_4(m-1)},$$

where s is the sample standard deviation and $c_4(\cdot)$ is the correction factor commonly tabulated in the SPC reference texts (see Montgomery (2020)).

Our RL procedure mirrors the conventional approach used in standard EWMA chart design. In this setting, the PPO agent with policy $\pi_\theta(\cdot)$ is trained under the known parameter setting (Case K), where the underlying process is distributed as $N(0, 1)$, exactly as described in Section 3.2. Then a decision threshold is calibrated to achieve a specified ARL_0 in this known environment, serving a function comparable to choosing control limits in the conventional EWMA design. To assess the effect of parameter estimation, we applied the calibrated threshold without modification to monitor the process in the unknown parameter setting (Case U). Each evaluation episode begins with a new Phase I reference sample, from which $(\hat{\mu}_0, \hat{\sigma}_0)$ are calculated and used to standardize the Phase II observations.

Tables 7 and 8 report the ARL, SDRL, $Q_{0.10}$, median, and $Q_{0.90}$ based on $K = 10{,}000$ independent Phase II trajectories for each configuration. Let $R_{(1)} \le \cdots \le R_{(K)}$ denote the ordered run lengths. The empirical quantile uses linear interpolation: for $h = 1 + (K-1)\alpha$, $j = \lfloor h \rfloor$, and $\gamma = h - j$,

$$Q_\alpha = (1-\gamma)R_{(j)} + \gamma R_{(j+1)}.$$

The median, $Q_{0.10}$, and $Q_{0.90}$ correspond to $\alpha = 0.50$, 0.10, and 0.90, respectively. The SDRL is reported to describe run-length variability. Because $K = 10{,}000$ for every configuration, the Monte Carlo standard error of an estimated ARL is $\mathrm{SDRL}/\sqrt{K} = \mathrm{SDRL}/100$. All Phase II runs were limited to 10,000 observations. Therefore, the recorded run length was $R_i^* = \min(R_i, 10{,}000)$, and the reported means and standard deviations are restricted run length statistics. In Table 8, $Q_{0.90}$ reaches the truncation limit for the IC configurations with $m = 20$ and $\lambda \in \{0.05, 0.14\}$. With $K = 10{,}000$ and the stated quantile convention, this indicates that at least 10.01% of the trajectories were right-censored. The corresponding $Q_{0.90}$ values are therefore reported as lower bounds and are not interpreted as exact upper-tail quantiles.. An important empirical observation is that the IC ARL_0 in Case U deviates from its target value, particularly when m is small. In addition, the IC run length distribution becomes more dispersed, as reflected by the increase in SDRL and the widening gap between $Q_{0.10}$ and $Q_{0.90}$.

For subgroup EWMA charts, Jones et al. (2001) showed that parameter estimation can, on average, lead to more frequent false alarms and simultaneously reduce OOC sensitivity. For individual EWMA charts, Human et al. (2011) stated that Case U performance can deviate from Case K in either direction, depending on the estimator and design, and that ARL alone can mask substantial distributional changes. In the proposed design, the calibrated $\eta^\star$ is fixed from Case K and is not retuned under estimation. Thus, any IC ARL_0 inflation observed in Case U can be interpreted as the policy becoming effectively more conservative under the state distribution induced by $(\hat{\mu}_0, \hat{\sigma}_0)$. This effect diminishes as m increases, consistent with previous studies showing that the run length distribution under estimated parameters approaches Case K behavior as the Phase I reference sample increases.

For both targets, Tables 8 and 7 indicate that, for a given

δ, the OOC $\mathrm{ARL}_1(\delta)$ in Case U is generally greater than the corresponding value in Case K (Tables 3 and 4). This pattern is consistent with earlier results for EWMA charts under Case U (Jones et al., 2001; Human et al., 2011). For the proposed chart, this corresponds to evaluating the decision rule induced by $(\pi_\theta, \eta^\star)$ under a different state distribution. In Case U, the standardized shift uses episode-specific $(\hat{\mu}_0, \hat{\sigma}_0)$, so the realized EWMA state s_t is subject to random mean distortion even when the underlying data remain normal. Keeping $\eta^\star$ fixed therefore changes the frequency with which trajectories enter regions where $\pi_\theta(1 \mid s_t) > \eta^\star$, affecting both false alarm behavior and OOC detection delay. From Tables 7 and 8, the following observations can be made:

- For Case U, the IC ARL_0 rises sharply when m is small, and the effect becomes more pronounced as the target ARL_0 increases. For the $\mathrm{ARL}_0 \approx 100$ design with $\lambda = 0.05$, the Case U IC mean is 991.98 at $m = 20$ and decreases to 350.11 at $m = 50$ and 184.36 at $m = 100$ (Table 7). A similar pattern is observed for other values of λ. For example, when $\lambda = 0.14$, the Case U IC mean is 537.25 ($m = 20$), 204.15 ($m = 50$), and 140.84 ($m = 100$). The pattern is also observed for $\mathrm{ARL}_0 \approx 500$. For $\lambda = 0.05$, the Case U IC mean is 2665.26 at $m = 20$, 1788.99 at $m = 50$, and 1254.55 at $m = 100$. For $\lambda = 0.70$, the corresponding means are 1411.66 ($m = 20$), 886.33 ($m = 50$), and 696.75 ($m = 100$) (Table 8). The observed increase in IC ARL_0 for Case U contrasts with the behavior reported for some classical EWMA designs, which tend to exhibit higher IC false alarm rates (Jones et al., 2001; Human et al., 2011).

- The OOC performance continues to degrade in Case U, but the extent of this degradation depends on δ and is most visible for small to moderate shifts. In the $\mathrm{ARL}_0 \approx 100$ design with $\lambda = 0.05$, $\mathrm{ARL}_1(0.50)$ is 16.70 at $m = 20$ and decreases to 13.42 at $m = 50$ and 12.67 at $m = 100$; meanwhile $\mathrm{ARL}_1(1.00)$ is 6.06 ($m = 20$), 5.86 ($m = 50$), and 5.84 ($m = 100$) (Table 7). For larger shifts the mean quickly approaches 1–2 across all m (e.g., at $\delta = 6.00$, means are 1.11, 1.08, and 1.07 for $m = 20, 50, 100$ when $\lambda = 0.05$), so the practical differences concentrate in the small to moderate shift regime that is also emphasized in classical EWMA comparisons.

- Small Phase I sample sizes and larger target ARL_0 values lead to longer upper-tail run lengths. For $\mathrm{ARL}_0 \approx 100$ and $\lambda = 0.05$, the IC $Q_{0.90}$ is 2792.50 at $m = 20$, compared with 661.10 at $m = 50$ and 397.00 at $m = 100$. For $\mathrm{ARL}_0 \approx 500$ and $\lambda = 0.05$, the IC $Q_{0.90}$ reaches 10000.00 at $m = 20$ and remains large at $m = 50$ (7993.60) and $m = 100$ (3500.50).

- The dependence on λ is present but secondary to the Phase I sample size m. At $\mathrm{ARL}_0 \approx 100$ and $m = 20$, ARL_0 decreases as λ increases from 0.05 to 0.70 (from 991.98 to 280.07), but it remains far above the target in all cases (Table 7). For $m = 100$, the IC means are closer across λ (for instance, 184.36 at $\lambda = 0.05$ versus 117.59 at $\lambda = 0.70$).

These results reflect the effect of applying a charting rule calibrated under known parameters to the estimated-parameter case without recalibration. In the classical EWMA setting, (λ, L) are selected under known parameters and then evaluated when the parameters are estimated. Similarly, in the proposed RL design, the threshold $\eta^\star$ is calibrated in Case K and then applied unchanged in Case U. In line with the literature, a large reference sample is required to attain performance comparable to the known-parameter case. Figure 3 compares the ARL profiles of the proposed chart under known and unknown parameters.

5.2. Conditional Expected Delay and Steady-State Performance

The results presented so far are based on the zero-state ARL. As defined in Section 3.4, CED offers an additional measure of detection performance for scenarios in which the process shift occurs only after the chart has been in operation for some time. The limiting value of CED_τ as τ grows large is the steady-state ARL (Kenett and Pollak, 2012). We present CED profiles over different change points and examine their behavior. A comprehensive treatment of CED and steady-state ARL performance for EWMA, CUSUM, and mixed EWMA–CUSUM charts is given by Ibrahim et al. (2025).

For this comparison, all charts were designed to attain a target $\mathrm{ARL}_0 \approx 100$. The smoothing parameter was set to $\lambda = 0.05$ for both the proposed chart and the classical EWMA chart, while the reference value for the CUSUM chart was set to $k = 0.25$. Alternative designs (e.g., $\lambda = 0.5$ for the RL-EWMA and EWMA charts and $k = 0.5$ for CUSUM) were also examined and yielded qualitatively similar results; these are therefore not reported here.

The CED profiles shown in Figure 4 reveal clear differences between the RL-EWMA chart and the classical charts. We summarize the main observations as follows:

- **Zero-state versus steady-state behavior:**

  The CED values of the RL-EWMA chart increase with τ before stabilizing. This may occur because the signaling decision depends jointly on X_t and Z_t. As τ increases, the calculation includes only runs with no earlier signal. Their states before the change tend to have lower signaling probabilities. More observations are therefore needed after the shift to gather enough evidence. The CED levels off when the distribution of these states stabilizes. The observed rise and subsequent stabilization are consistent with this conditional survival interpretation. The longer delay for later changes is a practical limitation when rapid steady-state detection is required. Differences between zero-state and steady-state performance have also been reported for other

Table 7: RL-EWMA chart performance under Case U for $ARL_0 \approx 100$.

		$m = 20$				$m = 50$				$m = 100$			
δ	Stat	$\lambda = 0.05$	0.14	0.50	0.70	$\lambda = 0.05$	0.14	0.50	0.70	$\lambda = 0.05$	0.14	0.50	0.70
0.00	ARL	991.98	537.25	294.99	280.07	350.11	204.15	151.50	142.78	184.36	140.84	124.53	117.59
	SDRL	2505.22	1608.26	874.79	820.23	1100.18	552.85	277.04	236.02	461.16	243.57	179.81	157.87
	$Q_{0.10}$	9.00	9.00	8.00	8.00	11.00	10.00	9.00	9.00	12.00	11.00	11.00	10.00
	Median	63.00	66.00	66.00	68.00	65.00	68.00	67.00	68.00	68.50	70.00	68.00	67.00
	$Q_{0.90}$	2792.50	1012.00	603.00	601.10	661.10	444.00	359.10	335.00	397.00	322.00	293.00	278.00
0.25	ARL	79.65	71.34	87.08	102.25	34.96	39.17	53.31	61.22	28.05	32.75	45.37	52.91
	SDRL	340.07	223.02	224.92	273.52	70.26	64.58	86.43	93.20	31.14	37.34	55.89	63.62
	$Q_{0.10}$	6.00	5.00	4.00	4.00	6.00	6.00	5.00	5.00	6.00	6.00	5.00	5.00
	Median	18.00	21.00	27.00	31.00	18.00	20.00	27.00	32.00	18.00	21.00	27.00	32.00
	$Q_{0.90}$	121.00	139.00	194.00	218.00	70.00	85.00	122.00	144.00	59.00	73.00	106.00	125.00
0.50	ARL	16.70	20.10	32.38	42.46	13.42	14.66	23.27	29.24	12.67	13.83	20.15	26.23
	SDRL	29.99	39.87	76.80	98.15	12.01	14.59	33.64	39.97	9.74	11.84	21.84	30.44
	$Q_{0.10}$	4.00	4.00	3.00	3.00	5.00	4.00	3.00	3.00	5.00	4.00	3.00	3.00
	Median	10.00	10.00	13.00	16.00	10.00	10.00	13.00	16.00	10.00	10.00	13.00	16.00
	$Q_{0.90}$	31.00	40.00	70.00	95.00	26.00	30.00	52.00	68.00	23.10	28.00	45.00	61.00
0.75	ARL	8.88	9.46	14.75	20.11	8.10	8.38	11.89	15.52	7.95	8.07	10.79	13.97
	SDRL	8.28	10.57	26.98	38.71	5.12	6.19	13.67	20.20	4.55	5.45	10.34	14.89
	$Q_{0.10}$	3.00	3.00	2.00	2.00	3.00	3.00	2.00	2.00	4.00	3.00	2.00	2.00
	Median	7.00	7.00	7.00	9.00	7.00	7.00	8.00	9.00	7.00	7.00	8.00	9.00
	$Q_{0.90}$	16.00	18.00	32.00	46.00	14.00	16.00	26.00	35.00	14.00	15.00	23.00	31.00
1.00	ARL	6.06	6.20	8.10	11.58	5.86	5.83	6.99	9.22	5.84	5.66	6.65	8.65
	SDRL	3.70	4.82	11.75	22.69	2.94	3.53	6.57	10.52	2.81	3.19	5.52	8.63
	$Q_{0.10}$	3.00	2.00	2.00	1.00	3.00	3.00	2.00	2.00	3.00	3.00	2.00	2.00
	Median	5.00	5.00	5.00	6.00	5.00	5.00	5.00	6.00	5.00	5.00	5.00	6.00
	$Q_{0.90}$	10.00	11.00	17.00	25.00	10.00	10.00	14.00	20.00	9.00	10.00	14.00	19.00
1.50	ARL	3.97	3.71	3.77	4.58	3.88	3.60	3.52	4.06	3.87	3.60	3.41	3.95
	SDRL	1.76	1.91	3.41	5.32	1.52	1.64	2.47	3.67	1.46	1.55	2.21	3.21
	$Q_{0.10}$	2.00	2.00	1.00	1.00	2.00	2.00	1.00	1.00	2.00	2.00	1.00	1.00
	Median	4.00	3.00	3.00	3.00	4.00	3.00	3.00	3.00	4.00	3.00	3.00	3.00
	$Q_{0.90}$	6.00	6.00	7.00	10.00	6.00	6.00	7.00	8.00	6.00	6.00	6.00	8.00
2.00	ARL	2.97	2.71	2.36	2.57	2.95	2.65	2.29	2.40	2.94	2.66	2.25	2.37
	SDRL	1.08	1.10	1.52	2.23	0.96	0.98	1.28	1.71	0.93	0.96	1.19	1.61
	$Q_{0.10}$	2.00	2.00	1.00	1.00	2.00	2.00	1.00	1.00	2.00	2.00	1.00	1.00
	Median	3.00	2.00	2.00	2.00	3.00	2.00	2.00	2.00	3.00	2.00	2.00	2.00
	$Q_{0.90}$	4.00	4.00	4.00	5.00	4.00	4.00	4.00	4.10	4.00	4.00	4.00	4.00
2.50	ARL	2.42	2.17	1.73	1.75	2.42	2.16	1.71	1.69	2.41	2.17	1.71	1.65
	SDRL	0.77	0.78	0.88	1.13	0.69	0.69	0.81	0.96	0.66	0.67	0.80	0.90
	$Q_{0.10}$	2.00	1.00	1.00	1.00	2.00	1.00	1.00	1.00	2.00	1.00	1.00	1.00
	Median	2.00	2.00	2.00	1.00	2.00	2.00	2.00	1.00	2.00	2.00	2.00	1.00
	$Q_{0.90}$	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00
3.00	ARL	2.08	1.85	1.43	1.37	2.07	1.85	1.40	1.34	2.08	1.83	1.39	1.33
	SDRL	0.60	0.62	0.62	0.68	0.53	0.58	0.58	0.62	0.51	0.56	0.57	0.58
	$Q_{0.10}$	1.00	1.00	1.00	1.00	2.00	1.00	1.00	1.00	2.00	1.00	1.00	1.00
	Median	2.00	2.00	1.00	1.00	2.00	2.00	1.00	1.00	2.00	2.00	1.00	1.00
	$Q_{0.90}$	3.00	3.00	2.00	2.00	3.00	2.00	2.00	2.00	3.00	2.00	2.00	2.00
4.00	ARL	1.65	1.42	1.11	1.08	1.67	1.39	1.09	1.06	1.66	1.39	1.09	1.06
	SDRL	0.51	0.51	0.32	0.29	0.50	0.50	0.29	0.24	0.49	0.49	0.29	0.24
	$Q_{0.10}$	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	Median	2.00	1.00	1.00	1.00	2.00	1.00	1.00	1.00	2.00	1.00	1.00	1.00
	$Q_{0.90}$	2.00	2.00	2.00	1.00	2.00	2.00	1.00	1.00	2.00	2.00	1.00	1.00

Table 8: RL-EWMA chart performance under Case U for $ARL_0 \approx 500$.

		$m = 20$				$m = 50$				$m = 100$			
δ	Stat	$\lambda = 0.05$	0.14	0.50	0.70	$\lambda = 0.05$	0.14	0.50	0.70	$\lambda = 0.05$	0.14	0.50	0.70
0.00	ARL	2665.26	2058.48	1630.13	1411.66	1788.99	1284.27	973.51	886.33	1254.55	884.84	699.85	696.75
	SDRL	3923.61	3377.29	2866.42	2571.83	3040.54	2350.32	1782.43	1593.07	2276.50	1570.34	1104.48	1078.84
	$Q_{0.10}$	24.00	22.00	24.00	25.00	36.00	33.00	35.00	36.00	46.00	43.00	41.00	44.00
	Median	322.50	312.00	323.00	310.00	331.50	323.00	320.00	315.00	343.00	343.00	330.00	336.00
	$Q_{0.90}$	≥ 10000	≥ 10000	6316.40	4583.30	7993.60	3833.20	2490.10	2216.40	3500.50	2159.00	1711.10	1689.20
0.25	ARL	500.65	462.94	596.54	671.33	122.67	154.84	271.00	331.70	82.07	104.68	193.27	252.25
	SDRL	1702.48	1482.57	1598.13	1673.84	366.06	402.36	599.84	693.46	133.28	164.84	295.87	380.10
	$Q_{0.10}$	12.00	10.00	10.00	11.00	14.00	12.00	14.00	15.00	15.00	13.00	15.00	17.00
	Median	46.00	55.00	97.00	121.00	46.00	56.00	99.00	122.00	46.00	55.00	99.00	128.00
	$Q_{0.90}$	671.30	801.10	1245.50	1537.10	232.00	326.00	619.10	790.10	169.00	238.00	460.00	607.10
0.50	ARL	44.32	64.63	165.51	248.00	27.99	34.85	78.73	128.81	25.04	30.00	66.48	101.39
	SDRL	121.84	210.72	559.48	810.64	33.96	51.59	145.35	251.74	19.21	34.65	94.33	139.70
	$Q_{0.10}$	8.00	6.00	5.00	6.00	9.00	7.00	6.00	7.00	9.00	7.00	7.00	9.00
	Median	20.00	20.00	34.00	51.00	20.00	20.00	37.00	54.00	20.00	20.00	38.00	57.00
	$Q_{0.90}$	75.00	119.00	331.00	478.10	52.00	73.00	180.00	298.00	45.00	61.00	152.00	236.00
0.75	ARL	17.26	20.04	53.53	103.55	14.69	15.37	32.41	55.51	14.26	14.25	27.29	45.70
	SDRL	23.43	37.74	149.57	316.83	9.06	14.52	52.71	93.50	7.50	10.54	31.70	61.42
	$Q_{0.10}$	6.00	5.00	3.00	4.00	7.00	5.00	4.00	4.00	7.00	5.00	4.00	5.00
	Median	12.00	11.00	17.00	24.00	13.00	11.00	17.00	27.00	13.00	11.00	17.00	26.00
	$Q_{0.90}$	30.00	38.00	110.10	214.00	25.00	29.00	74.00	129.00	24.00	27.00	62.00	107.00
1.00	ARL	10.65	10.63	22.00	43.46	9.98	9.31	15.14	26.95	9.97	8.92	13.93	22.92
	SDRL	6.60	10.51	49.22	113.56	4.62	6.17	18.22	42.92	4.27	5.01	14.36	27.53
	$Q_{0.10}$	5.00	4.00	3.00	3.00	5.00	4.00	3.00	3.00	6.00	4.00	3.00	3.00
	Median	9.00	8.00	9.00	14.00	9.00	8.00	9.00	14.00	9.00	8.00	9.00	14.00
	$Q_{0.90}$	18.00	20.00	44.00	94.10	16.00	16.00	33.00	61.00	16.00	15.00	30.00	53.00
1.50	ARL	6.44	5.43	6.89	11.86	6.29	5.19	5.71	8.63	6.25	5.15	5.49	7.84
	SDRL	2.68	2.99	9.62	25.38	2.14	2.32	4.73	10.54	1.99	2.16	4.14	7.56
	$Q_{0.10}$	4.00	3.00	2.00	2.00	4.00	3.00	2.00	2.00	4.00	3.00	2.00	2.00
	Median	6.00	5.00	4.00	5.00	6.00	5.00	4.00	5.00	6.00	5.00	4.00	5.00
	$Q_{0.90}$	10.00	9.00	14.00	25.00	9.00	8.00	11.00	19.00	9.00	8.00	10.00	17.00
2.00	ARL	4.67	3.73	3.50	4.70	4.65	3.67	3.29	3.96	4.61	3.68	3.23	3.84
	SDRL	1.52	1.51	2.86	6.46	1.34	1.30	2.11	3.75	1.24	1.24	1.86	3.13
	$Q_{0.10}$	3.00	2.00	1.00	1.00	3.00	2.00	1.00	1.00	3.00	2.00	1.00	1.00
	Median	4.00	3.00	3.00	3.00	4.00	3.00	3.00	3.00	4.00	3.00	3.00	3.00
	$Q_{0.90}$	7.00	6.00	6.00	10.00	6.00	5.00	6.00	8.00	6.00	5.00	6.00	8.00
2.50	ARL	3.73	2.92	2.37	2.67	3.71	2.91	2.28	2.41	3.70	2.89	2.26	2.36
	SDRL	1.09	1.00	1.42	2.73	0.93	0.88	1.15	1.67	0.89	0.85	1.06	1.54
	$Q_{0.10}$	3.00	2.00	1.00	1.00	3.00	2.00	1.00	1.00	3.00	2.00	1.00	1.00
	Median	4.00	3.00	2.00	2.00	4.00	3.00	2.00	2.00	4.00	3.00	2.00	2.00
	$Q_{0.90}$	5.00	4.00	4.00	5.00	5.00	4.00	4.00	4.00	5.00	4.00	4.00	4.00
3.00	ARL	3.13	2.44	1.80	1.83	3.12	2.42	1.78	1.71	3.13	2.40	1.77	1.69
	SDRL	0.83	0.74	0.86	1.21	0.73	0.65	0.78	0.96	0.70	0.62	0.74	0.90
	$Q_{0.10}$	2.00	2.00	1.00	1.00	2.00	2.00	1.00	1.00	2.00	2.00	1.00	1.00
	Median	3.00	2.00	2.00	2.00	3.00	2.00	2.00	2.00	3.00	2.00	2.00	2.00
	$Q_{0.90}$	4.00	3.00	3.00	3.00	4.00	3.00	3.00	3.00	4.00	3.00	3.00	3.00
4.00	ARL	2.41	1.89	1.29	1.21	2.40	1.91	1.26	1.18	2.38	1.91	1.25	1.17
	SDRL	0.57	0.52	0.49	0.47	0.53	0.46	0.46	0.41	0.50	0.43	0.45	0.41
	$Q_{0.10}$	2.00	1.00	1.00	1.00	2.00	1.00	1.00	1.00	2.00	1.00	1.00	1.00
	Median	2.00	2.00	1.00	1.00	2.00	2.00	1.00	1.00	2.00	2.00	1.00	1.00
	$Q_{0.90}$	3.00	2.00	2.00	2.00	3.00	2.00	2.00	2.00	3.00	2.00	2.00	2.00

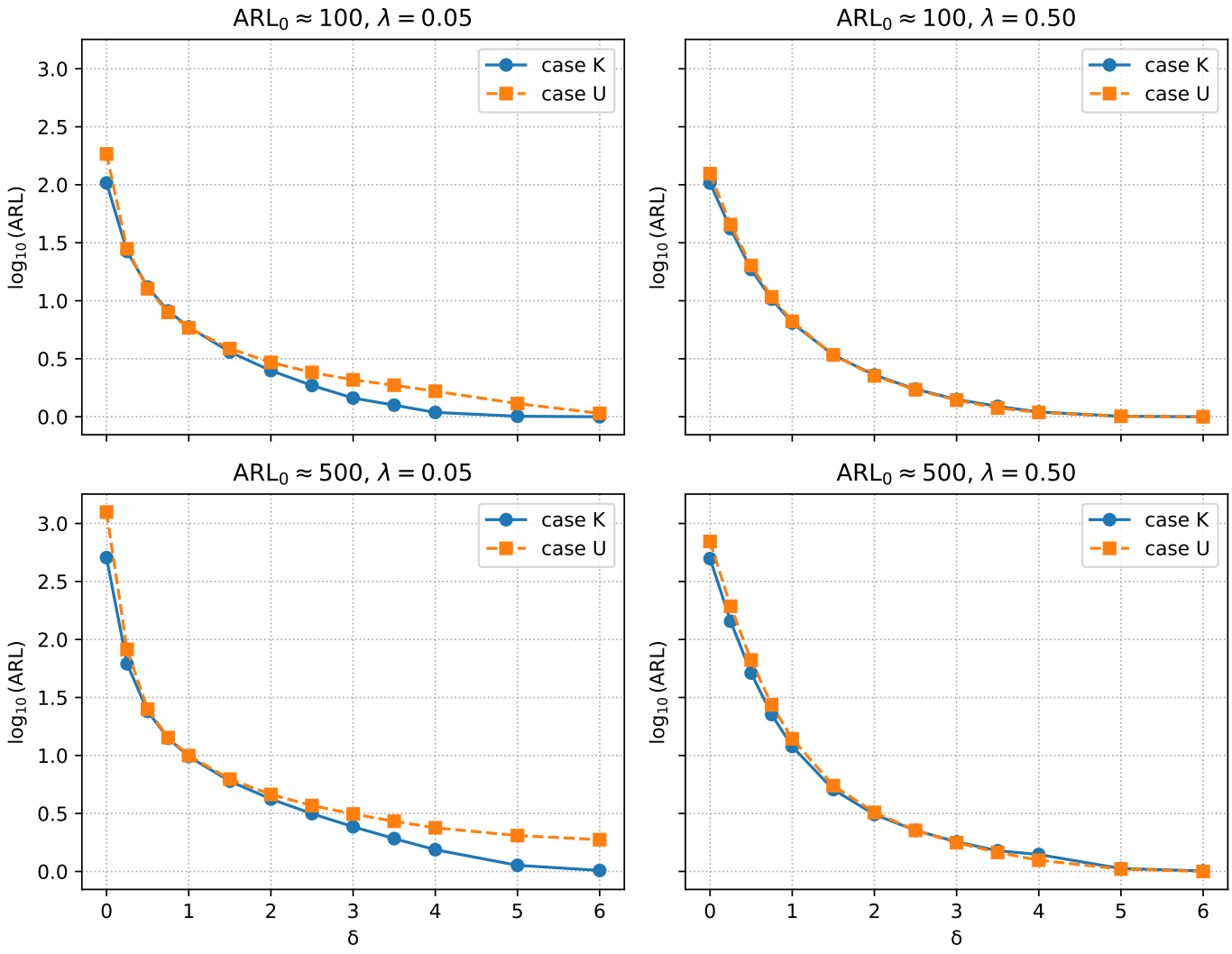


Figure 3: ARL comparison for known and unknown parameter cases for selected values of $\lambda \in \{0.05, 0.50\}$ and $m = 100$.

memory-type control charts (Alevizakos et al., 2025). Despite this increase, the RL-EWMA chart maintains lower CED values than the classical EWMA and CUSUM charts for most of the evaluated shifts and change points.

- At $\delta = 0.25$, the performance advantage of the RL-EWMA chart is more pronounced. It stabilizes at a steady-state CED between approximately 25 and 28, whereas the classical EWMA and CUSUM charts level off at approximately 34 and 36, respectively.
- At $\delta = 0.50$, the steady-state CED based on RL is approximately 14, outperforming CUSUM ($\approx$ 15) and EWMA ($\approx$ 16).
- At $\delta = 1.00$, the RL-EWMA chart maintains a steady-state delay of approximately 6.7, compared to 7.4 for the classical EWMA.
- At $\delta = 2.0$, the RL-EWMA chart maintains its advantage, with a steady-state CED of approximately 2.9, while CUSUM and EWMA stabilize at 3.0 and 3.75, respectively.

These results indicate that the signaling policy learned by the PPO agent is more efficient for continuous process monitoring than traditional control charts. In addition to the enhanced performance reported in Section 4, the proposed chart also delivers superior steady-state behavior over a broad spectrum of shift magnitudes.

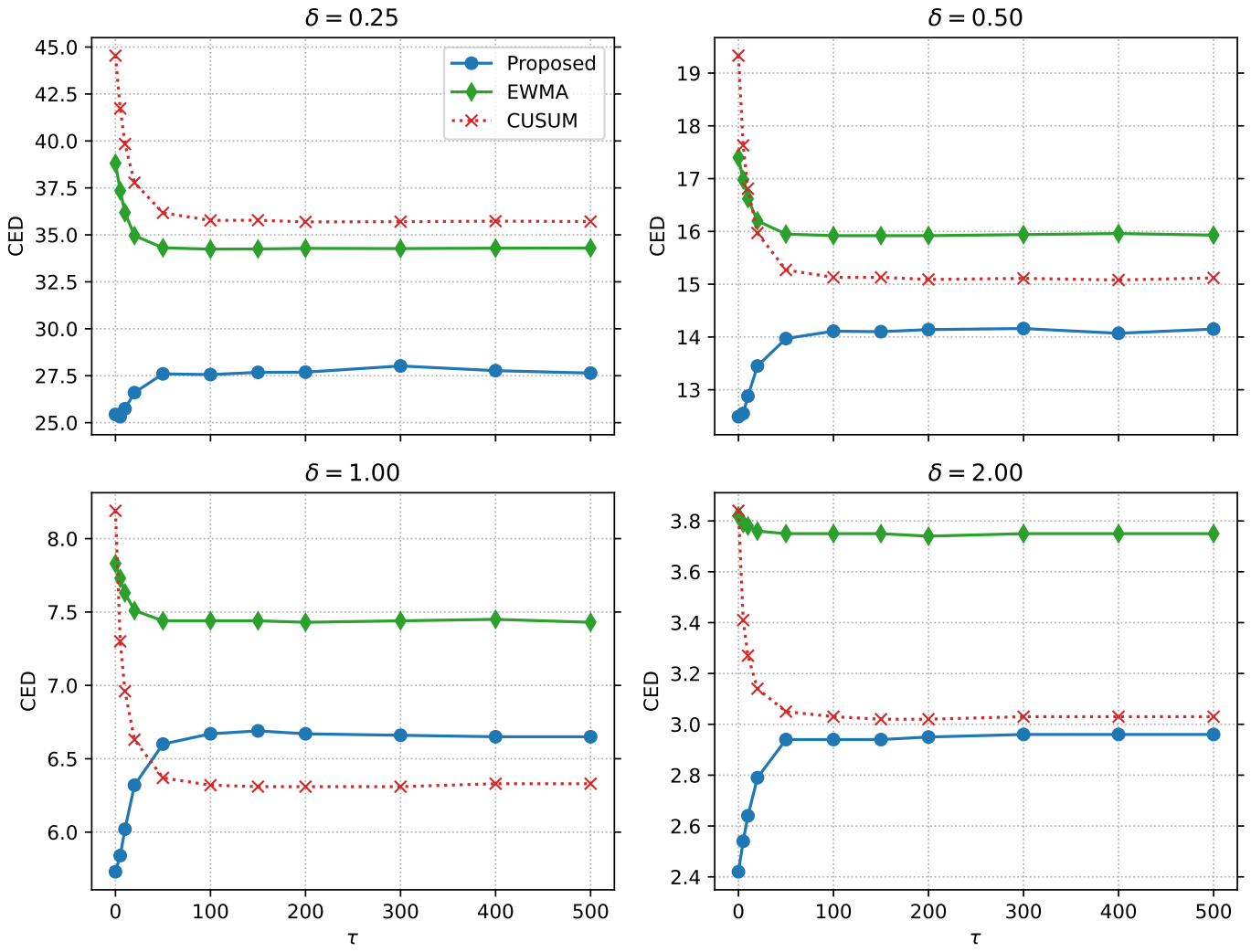


Figure 4: CED versus change point τ for the RL-EWMA, classical EWMA, and CUSUM charts across various shift magnitudes δ.

5.3. *The Normality Assumption and Possible Extensions to Nonnormal Processes*

We assume a Gaussian IC process for both the training and evaluation phases. This choice follows the classical literature on control charts. Maintaining this assumption allows the effect of Phase I parameter estimation to be examined in isolation, without confounding distributional misspecification. The robustness of EWMA charts to nonnormal distributions has been investigated extensively. For example, Human et al. (2011) and Borror et al. (1999) showed that EWMA charts with small smoothing constants can exhibit reasonable robustness to symmetric heavy-tailed and mildly skewed distributions when parameters are known. This robustness is one reason why EWMA charts are often preferred over Shewhart-type charts in nonnormal settings.

Nonnormality constitutes another type of distribution shift beyond Phase I parameter estimation. Because the policy π_θ is trained under a Gaussian IC distribution, departures from normality can alter the frequency with which the observed state enters regions associated with signaling. The standardized t_{10} experiment provides a limited sensitivity check and is not intended to establish general robustness under heavy-tailed, skewed, or contaminated distributions. A systematic analysis of these distributions is left for future research.

6. Urban Mobility Monitoring Using CTA Bus Ridership

Monthly bus ridership provides an observable measure of public transportation demand and can therefore support the monitoring of changes in urban mobility patterns.

6.1. The Data

The case study used monthly bus ridership records from the Chicago Transit Authority (CTA). The data were obtained from the Chicago City Open Data Portal (Chicago Transit Authority, b). The dataset comprises monthly aggregate ridership counts and monthly day-type average values for each bus route. The time series extends over more than two decades, spanning the period from January 2001 to November 2025. The records show periods of relatively stable ridership as well as pronounced variation.

For the case study, we focus on total ridership. This choice yields a meaningful, directly interpretable univariate time series that is widely used in transit demand analysis (Boisjoly et al., 2018). Nevertheless, the proposed chart is not limited to bus ridership. The same approach can be extended to other monitoring contexts, including manufacturing quality characteristics, equipment condition, and healthcare. The CTA data are used only as an empirical illustration of how the charts respond to observed ridership variation. The application is not intended to validate the method under nonnormality, autocorrelation, or heteroscedasticity, and no claim is made that the nominal ARL properties obtained under independent Gaussian observations apply to these data. Historical events are used only as context and not as independently validated change points.

6.2. Monitoring Scenarios

To examine different application contexts, we consider two monitoring scenarios. Scenario A focuses on a single route, while Scenario B addresses system-level monitoring using all available routes. All charts are tuned for comparable IC performance, targeting $\mathrm{ARL}_0 \approx 500$. For the EWMA chart, the smoothing parameter is $\lambda = 0.50$ with control limit factor $L = 3.0701$, while the CUSUM chart uses a reference value $k = 0.5$ and decision interval $h = 5.0707$. The proposed chart also uses $\lambda = 0.50$ and converts the policy output into a signal using a probability threshold. We calibrated the two scenarios separately using the bisection procedure in Section 3.2. Both calibrations targeted $\mathrm{ARL}_0 = 500$. The resulting thresholds were approximately $\eta = 0.25$ for route-level monitoring and $\eta = 0.50$ for aggregate monitoring.

6.2.1. Scenario A: Route-Level Monitoring (Route 8)

In the first scenario, we examine a single route (Route 8 Halsted), which serves as a corridor-level monitoring task. The CTA Route 8 Halsted bus is a major north–south transit route in Chicago and provides extended daily service for commuters with diverse schedules. The route runs between Halsted/79th and Broadway/Waveland, with weekday service from 4:05 a.m. to 12:30 a.m. according to the Chicago Transit Authority (Chicago Transit Authority, a). Its frequent service, strong ridership, and susceptibility to localized disturbances make Route 8 well suited for corridor-level demand monitoring.



Figure 5: CTA bus in service on Halsted Street in Chicago, Illinois. Source: Chicago Transit Authority.

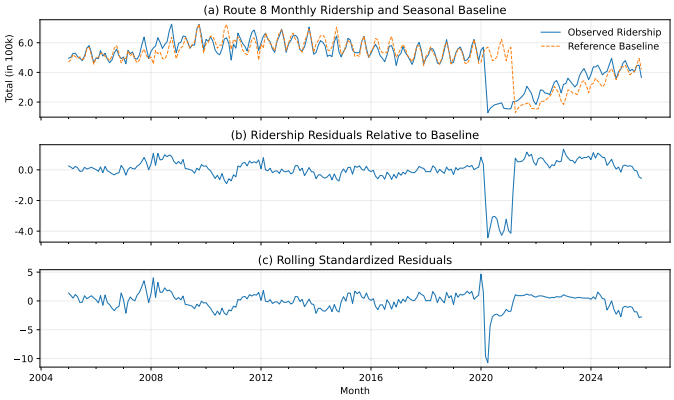


Figure 6: Monthly ridership for CTA Route 8 and successive stages of preprocessing: (a) observed ridership with seasonal reference baseline, (b) deviations from the baseline, and (c) rolling standardized residuals used as input for monitoring.

Figure 6 shows the total monthly ridership observed for Route 8. The series exhibits strong seasonality, gradual changes over time, and pronounced declines in observed ridership. A map showing a portion of Route 8 is provided in Figure 7 to give spatial context for the route.

6.2.2. Scenario B: System-Level Monitoring (Aggregate Average per Route)

In the second scenario, we examine monitoring at the system level. Simply aggregating ridership by summing trips across all routes can be misleading, because new routes are periodically introduced while others are removed at different times. As a result, variations in the aggregate may primarily reflect changes in network coverage rather than changes in underlying demand. To address this issue, we construct an aggregate series defined as

$$x_t = \frac{\sum_r \texttt{MonthlyTotal}_{r,t}}{N_t}, \tag{12}$$

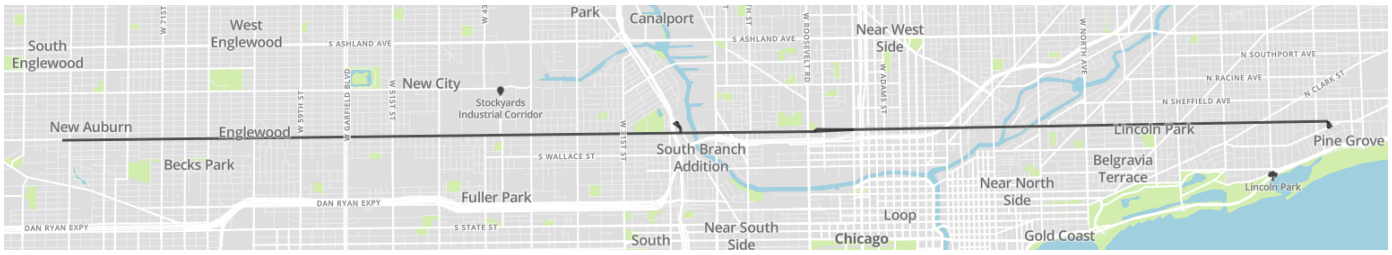


Figure 7: Map segment of CTA Route 8, Halsted.

where N_t denotes the number of routes that report data in month t. This normalization produces a quantity that can be interpreted as the average monthly ridership per active route.

The corresponding aggregate time series is presented in Figure 8. This scenario represents a system-level monitoring application that focuses on tracking overall demand patterns while maintaining robustness to route-level turnover.

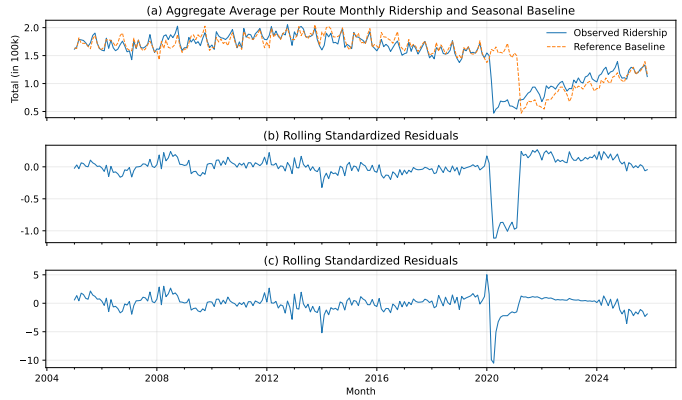


Figure 8: Monthly aggregate average ridership per active route and successive stages of preprocessing: (a) observed ridership with seasonal reference baseline, (b) baseline-adjusted deviations, and (c) rolling standardized residuals used as input for monitoring.

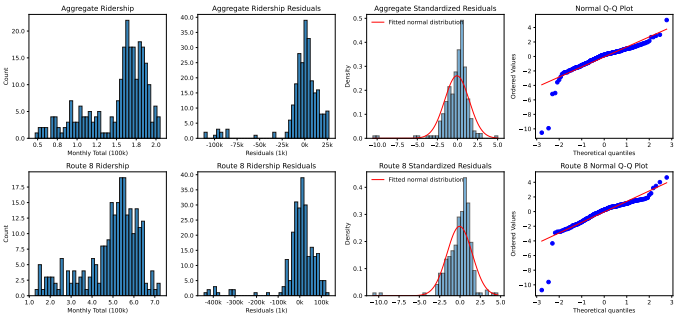


Figure 9: Aggregate and route-level ridership distributions after baseline adjustment: distributions of raw ridership, baseline-adjusted residuals, and rolling standardized residuals for the aggregate average per route series (top) and Route 8 (bottom), together with normal quantile–quantile (Q–Q) plots.

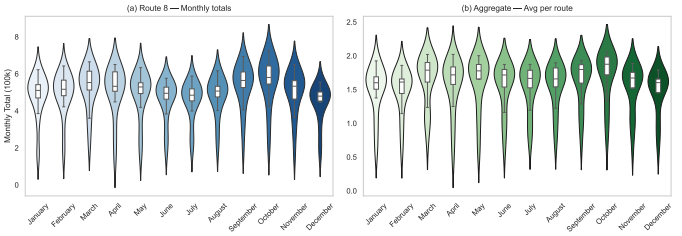


Figure 10: Month-of-year violin plots for Route 8 (a) and the aggregate series (b).

6.3. *Data Exploration and Preprocessing*

The exploratory plots suggest that both time series exhibit seasonality and nonstationarity. The histograms in Figure 9 are skewed with heavy tails, and the month-of-year violin plots (Figure 10) show systematic intra-annual variation. Therefore, we conduct monitoring on the residuals rather than on the raw counts. A similar strategy has been adopted in previous work (see, for instance, (Zeroual et al., 2017)).

For each series x_t, we form a baseline forecast $\hat{x}_t$ using a seasonal naïve model:

$$\hat{x}_t = x_{t-12}, \tag{13}$$

with a fallback to a 36-month trailing mean (shifted by one period) when seasonal lags are unavailable. Residuals are defined as

$$e_t = x_t - \hat{x}_t. \tag{14}$$

Figures 6 and 8 show the raw ridership series and preprocessing stages for Route 8 and the overall system, respectively. Since the residual variance is time-varying, we apply a trailing rolling z-score with a window of $w = 36$ for standardization:

$$z_t = \frac{e_t - \mu_{t-1}^{(w)}}{\sigma_{t-1}^{(w)}}, \tag{15}$$

where $\mu_{t-1}^{(w)}$ and $\sigma_{t-1}^{(w)}$ are computed from the past residuals only. All preprocessing was performed sequentially using only information available before month t.

The Shapiro–Wilk test rejected normality ($p < 0.01$), indicating some departure from the Gaussian distribution after preprocessing. However, the standardized aggregate residuals were not rejected against a standardized t_{10} distribution by the Kolmogorov–Smirnov test ($p > 0.05$). As a robustness check for this application, the trained policy and the threshold calibrated under the Gaussian assumption were evaluated using standardized t_{10} observations. The resulting ARL values are reported in Table B.10. The difference between the ARL values for the t_{10} and Gaussian cases is less than 5%. This result applies only to the standardized t_{10} setting examined here and is not interpreted as evidence of robustness across other nonnormal distributions.

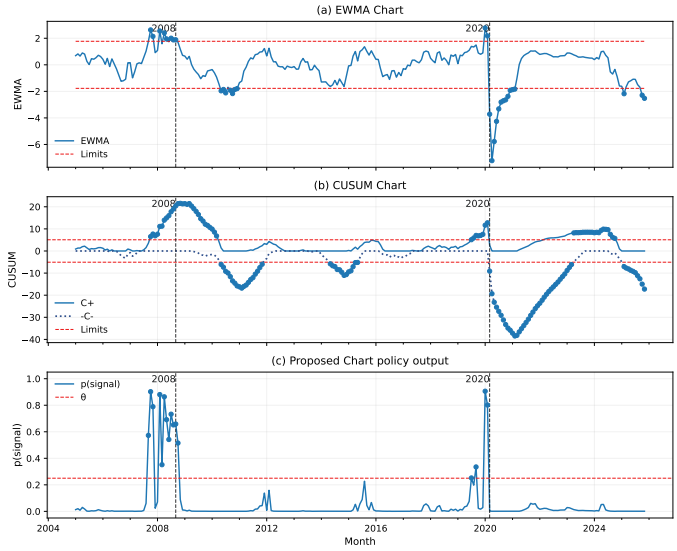


Figure 11: Monitoring results for CTA Route 8: (a) EWMA chart, (b) CUSUM chart, (c) proposed RL-EWMA chart.

Figure 11 presents the monitoring results for CTA Route 8 using the EWMA chart, the CUSUM chart, and the proposed RL-EWMA chart applied to the standardized residual series.

Figure 12 presents another set of monitoring results (Scenario B) obtained by applying the three competing charts to the aggregate average per route series.

All three charts respond to the pronounced changes observed in 2007–2008 and 2020, but their signaling patterns differ in timing and persistence. Because the change points are not independently labeled and the observations are standardized using rolling estimates, these plots only illustrate the behavior of the charts. Smaller signals outside these periods may reflect transient local changes or false alarms. Without additional information, these signals cannot be classified conclusively.

6.4. Managerial Insights

The proposed RL-EWMA chart supports a straightforward operational policy. At each sampling time, the choice is simply whether to continue monitoring or to trigger a signal for further investigation. This binary structure is practical because managers can interpret the chart output directly.

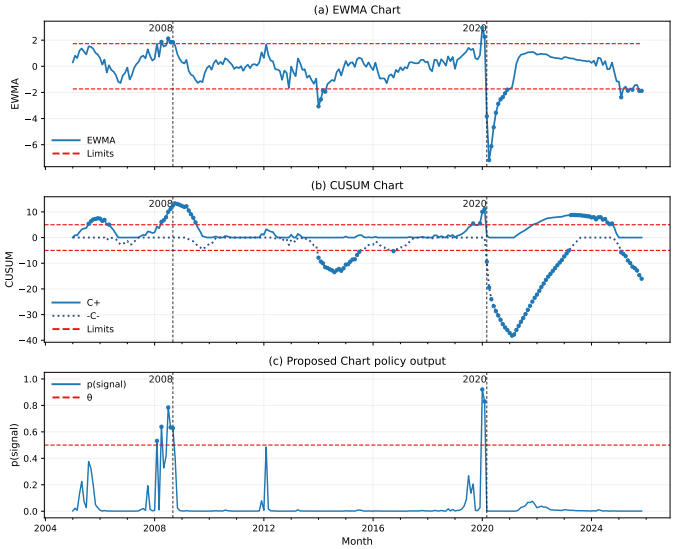


Figure 12: Monitoring results for the system-level aggregate (average monthly ridership per active route): (a) EWMA chart, (b) CUSUM chart, (c) proposed RL-EWMA chart.

According to the simulation study, the proposed chart performs especially well for small to moderate shifts in the process mean under both $\mathrm{ARL}_0 \approx 100$ and $\mathrm{ARL}_0 \approx 500$. This is reflected in the lower ARL_1 and EQL values shown in Tables 3 and 4. These results suggest that the learned policy is more efficient when the timely detection of modest changes is a primary objective.

In the CTA ridership case study, the proposed chart issues signals around major disruption episodes, most notably the 2007–2008 decline and the 2020 COVID-19 shock.

7. Conclusion

This study proposes an RL-based EWMA control chart in which PPO learns a nonlinear signaling probability from the observed state for efficient process monitoring. We then tune the threshold η to convert this probability into a stopping decision and achieve the target ARL_0. The performance of the chart is assessed using ARL, EARL, EQL, and CED.

Comparisons with adaptive EWMA, classical EWMA, and CUSUM charts show that the proposed method delivers better ARL_1 performance for small to moderate shifts, along with reduced EQL values. These findings indicate that learning a data-driven signaling strategy can yield faster detection than using classical control charts.

We also present an urban mobility monitoring application using real-world transit ridership data. The chart issues signals around pronounced variations in the time series. The case study illustrates the response of the chart when applied to residuals standardized using rolling estimates.

We adopted some simplifying assumptions. In particular, the process distribution was assumed to be normal with known parameters, shifts were modeled as permanent changes in the mean, and the EWMA smoothing parameter was fixed during training and evaluation. We also examined how parameter estimation, when parameters are unknown, influences the results. Future work can generalize the proposed chart in multiple ways. Possible extensions include non-Gaussian and multivariate monitoring. For example, the state could combine the current observation vector with a multivariate EWMA statistic, allowing the agent to learn a signaling rule that accounts for dependence among several process variables. The RL agent could also jointly adapt the signaling rule and chart parameters. Examining alternative reward formulations and online learning schemes can further clarify how RL can be incorporated more broadly into SPC. Future research can also examine economic designs in which cost parameters directly determine ARL values, removing the need for post-training calibration.

Appendix A. PPO hyperparameters and training settings

Table A.9 summarizes the PPO hyperparameters, environment configuration, and reward settings used to train and evaluate the proposed RL-EWMA control chart.

Appendix B. Sensitivity to nonnormality

The results are reported in Table B.10.

Appendix C. PPO training logs

Figure C.13 presents the PPO training logs and optimization diagnostics used to assess the stability of the learning process.

Appendix D. PPO training details

The PPO agent was trained on a Windows 11 computer with an Intel processor, 16 GB RAM, and no GPU. The implementation used Python 3.12.3, PyTorch 2.7.0, and Stable-Baselines3 2.3.0. Eight parallel environments were used. The policy had two hidden layers with 64 neurons each and 8,899 trainable parameters. Training for 200,000 steps took about 2,500 seconds.

Table A.9: PPO training, environment, and reward settings used for the RL-EWMA control chart.

Item	Value
RL library / policy	Stable-Baselines3 2.3.0, PPO with `MlpPolicy`
Learning rate	3×10^{-4}
Discount factor γ	0.99
GAE parameter λ_{GAE}	0.95
PPO clip range ϵ	0.2
Value function clip range	0.1
Entropy coefficient	1×10^{-5}
Batch size	128
n_steps	2048
PPO epochs per update (n_epochs)	10
Number of environments	8
Total timesteps	2×10^5
Observation normalization	`VecNormalize`: norm_obs=True, norm_reward=False, clip_obs=10.0
IC distribution	$X_t \sim \mathcal{N}(0,1)$
EWMA statistic	$Z_t = \lambda X_t + (1-\lambda)Z_{t-1}$, $Z_0 = \mu_0$; observation is (X_t, Z_t)
Action space	0=`WAIT` (continue monitoring), 1=`SIGNAL` (raise an alarm)
Training shift injection	Geometric, $p = 0.02$ per step
Training shift magnitude	$\delta \sim \mathcal{U}(0.50, 3.00)$
Episode truncation	maximum number of steps, `MAX_STEPS`=10,000
Reward: delay cost c_d	0.20
Reward: signal reward R_{sig}	1
c_s	0
Penalty for false alarm P_{FA}	1
Unlisted PPO parameters	Stable-Baselines3 2.3.0 defaults



Figure C.13: PPO training logs for the RL-EWMA control chart.

References

Abbas, N. (2023). On efficient change point detection using a step cumulative sum control chart. *Quality Engineering*, 35(4):712–728.

Abbas, T., Albogamy, F. R., and Abid, M. (2025). A machine learning approach to adaptive ewma control charts: Insights

Table B.10: ARL performance of the RL-EWMA chart under a standardized t_{10} distribution.

Distribution	ARL												
	0.00	0.25	0.50	0.75	1.00	1.50	2.00	2.50	3.00	3.50	4.00	5.00	6.00
Standardized t_{10}	487.54	149.10	53.19	23.65	12.24	5.27	3.20	2.24	1.75	1.50	1.38	1.04	1.00

from cardiac surgery data. *Quality and Reliability Engineering International*, 41(6):2567–2575.

Ajab, M., Zaman, B., Mukhtiar, F., Butt, N. R., and Faraz, M. I. (2026). Enhancing statistical process control with machine learning: The iso-cusum control chart for multivariate process. *Computers & Industrial Engineering*, 212:111750.

Alevizakos, V., Chatterjee, K., and Koukouvinos, C. (2025). On the performance and comparison of various memory-type control charts. *Communications in Statistics - Simulation and Computation*, 54(6):2089–2109.

Atalay, M., Testik, M. C., Duran, S., and Weiß, C. H. (2020). Guidelines for automating phase i of control charts by considering effects on phase-ii performance of individuals control chart. *Quality Engineering*, 32(2):223–243.

Boisjoly, G., Grisé, E., Maguire, M., Veillette, M.-P., Deboosere, R., Berrebi, E., and El-Geneidy, A. (2018). Invest in the ride: A 14 year longitudinal analysis of the determinants of public transport ridership in 25 north american cities. *Transportation Research Part A: Policy and Practice*, 116:434–445.

Borror, C. M., Montgomery, D. C., and Runger, G. C. (1999). Robustness of the ewma control chart to non-normality. *Journal of quality technology*, 31(3):309–316.

Capizzi, G. and Masarotto, G. (2003). An adaptive exponentially weighted moving average control chart. *Technometrics*, 45(3):199–207.

Castagliola, P., Maravelakis, P. E., and Figueiredo, F. O. (2016). The ewma median chart with estimated parameters. *IIE Transactions*, 48(1):66–74.

Chang, K. and Park, S. H. (2025). Reinforcement learning-based real-time contrasts control chart using an adaptive window size. *IEEE Access*, 13:72077–72086.

Chang, S. I., Ghafariasl, P., and Aryal, B. (2026). A review of artificial intelligence impacting statistical process monitoring and future directions. *Computers & Industrial Engineering*, 216:111989.

Chen, A. and Chen, Y. K. (2007). Design of ewma and cusum control charts subject to random shift sizes and quality impacts. *IIE Transactions*, 39(12):1127–1141.

Chicago Transit Authority. 8 Halsted (Bus Route Info). Accessed 17 January 2026.

Chicago Transit Authority. Cta - ridership - bus routes - monthly day-type averages & totals. Accessed 17 January 2026.

Colosimo, B. M., Jones-Farmer, L. A., Megahed, F. M., Paynabar, K., Ranjan, C., and Woodall, W. H. (2024). Statistical process monitoring from industry 2.0 to industry 4.0: Insights into research and practice. *Technometrics*, 66(4):507–530.

Ghasemian, P. and Noorossana, R. (2024). The inertial properties of ewma control charts. *Communications in Statistics - Theory and Methods*, 53(12):4542–4555.

Hightower, A., Ziedan, A., Guo, J., Zhu, X., and Brakewood, C. (2024). A comparison of time series methods for post-covid transit ridership forecasting. *Journal of Public Transportation*, 26:100097.

Huang, W., Shu, L., and Jiang, W. (2016). A gradient approach to the optimal design of cusum charts under unknown mean-shift sizes. *Journal of Quality Technology*, 48(1):68–83.

Human, S. W., Kritzinger, P., and Chakraborti, S. (2011). Robustness of the ewma control chart for individual observations. *Journal of Applied Statistics*, 38(10):2071–2087.

Ibrahim, U., Abbas, N., Riaz, M., and Mahmood, T. (2025). Smart monitoring solutions for real-time water ph regulation in aquatic ecotoxicology. *Journal of Chemometrics*, 39(4):e70024.

Jabeen, R., Zaka, A., and Khan, K. I. (2022). Classical estimator based modified control charts for phase-ii monitoring in real life. *Quality and Reliability Engineering International*, 38(5):2862–2880.

Jensen, W. A., Jones-Farmer, L. A., Champ, C. W., and Woodall, W. H. (2006). Effects of parameter estimation on control chart properties: a literature review. *Journal of Quality technology*, 38(4):349–364.

Jones, L. A., Champ, C. W., and Rigdon, S. E. (2001). The performance of exponentially weighted moving average charts with estimated parameters. *Technometrics*, 43(2):156–167.

Kazmi, M. W. and Noor-ul Amin, M. (2025). Integrating machine learning based ewma control charts for multivariate process monitoring. *Computers & Industrial Engineering*, 204:111131.

Kenett, R. S. and Pollak, M. (2012). On assessing the performance of sequential procedures for detecting a change. *Quality and Reliability Engineering International*, 28(5):500–507.

Knoth, S. (2025). *spc: Statistical Process Control – Calculation of ARL and Other Control Chart Performance Measures*. Comprehensive R Archive Network. R package version 0.7.2; accessed July 26, 2026.

Li, H., Ye, H., Cheng, J.-R. C., and Liu, K. (2025). Online monitoring of heterogeneous partially observable data streams based on q-learning. *IEEE Transactions on Automation Science and Engineering*, 22:4802–4817.

Montgomery, D. C. (2020). *Introduction to statistical quality control*. John Wiley & Sons.

PAGE, E. S. (1954). Continuous inspection schemes. *Biometrika*, 41(1-2):100–115.

Psarakis, S. (2015). Adaptive control charts: Recent developments and extensions. *Quality and Reliability Engineering International*, 31(7):1265–1280.

Rahimi, S. B., Shadman, A., and Harati, A. (2026). A model–free reinforcement learning framework for statistical process monitoring with variable sampling rate. *Engineering Applications of Artificial Intelligence*, 184:116311.

Roberts, S. (1959). Control chart tests based on geometric moving averages. *Technometrics*, 1(3):239–250.

Sabahno, H. (2027). A deep reinforcement learning framework for adaptive control charts. *Expert Systems with Applications*, 334:134414.

Schulman, J., Moritz, P., Levine, S., Jordan, M., and Abbeel, P. (2018). High-dimensional continuous control using generalized advantage estimation.

Schulman, J., Wolski, F., Dhariwal, P., Radford, A., and Klimov, O. (2017). Proximal policy optimization algorithms.

Shewhart, W. A. (1930). Economic quality control of manufactured product. *Bell System Technical Journal*, 9(2):364–389.

Sutton, R. S. and Barto, A. G. (2018). *Reinforcement learning: An introduction*. MIT press Cambridge.

Viharos, Z. J. and Jakab, R. (2021). Reinforcement learning for statistical process control in manufacturing. *Measurement*, 182:109616.

Zaman, B., Riaz, M., and Butt, N. R. (2025). Memory control chart based on machine learning technique for efficient process monitoring. *Computers & Industrial Engineering*, 201:110894.

Zeroual, A., Harrou, F., Sun, Y., and Messai, N. (2017). Monitoring road traffic congestion using a macroscopic traffic model and a statistical monitoring scheme. *Sustainable Cities and Society*, 35:494–510.